

Internet of Things

Lab 4



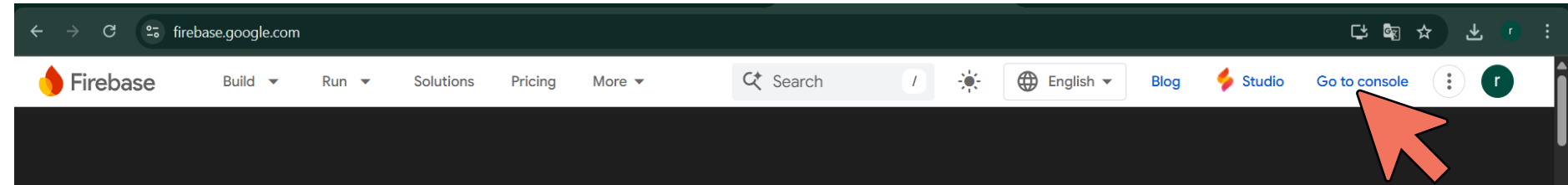
Firebase with IoT

- ❖ **Firebase** is Google's mobile and web application development platform that includes many services to manage data from IOS, Android, or web applications.
- ❖ This includes things like analytics, authentication, databases, configuration, file storage, push messaging, and the list goes on.
- ❖ The services are hosted in the cloud and scale with little to no effort on the part of the developer.
- ❖ You can use the ESP32 to connect and interact with your Firebase project, and you can create applications to control the ESP32 via Firebase from anywhere in the world.



Firebase: Creating a New Project

❖ Go to <https://firebase.google.com>, and then click on **Go to console**



Introducing Firebase Studio

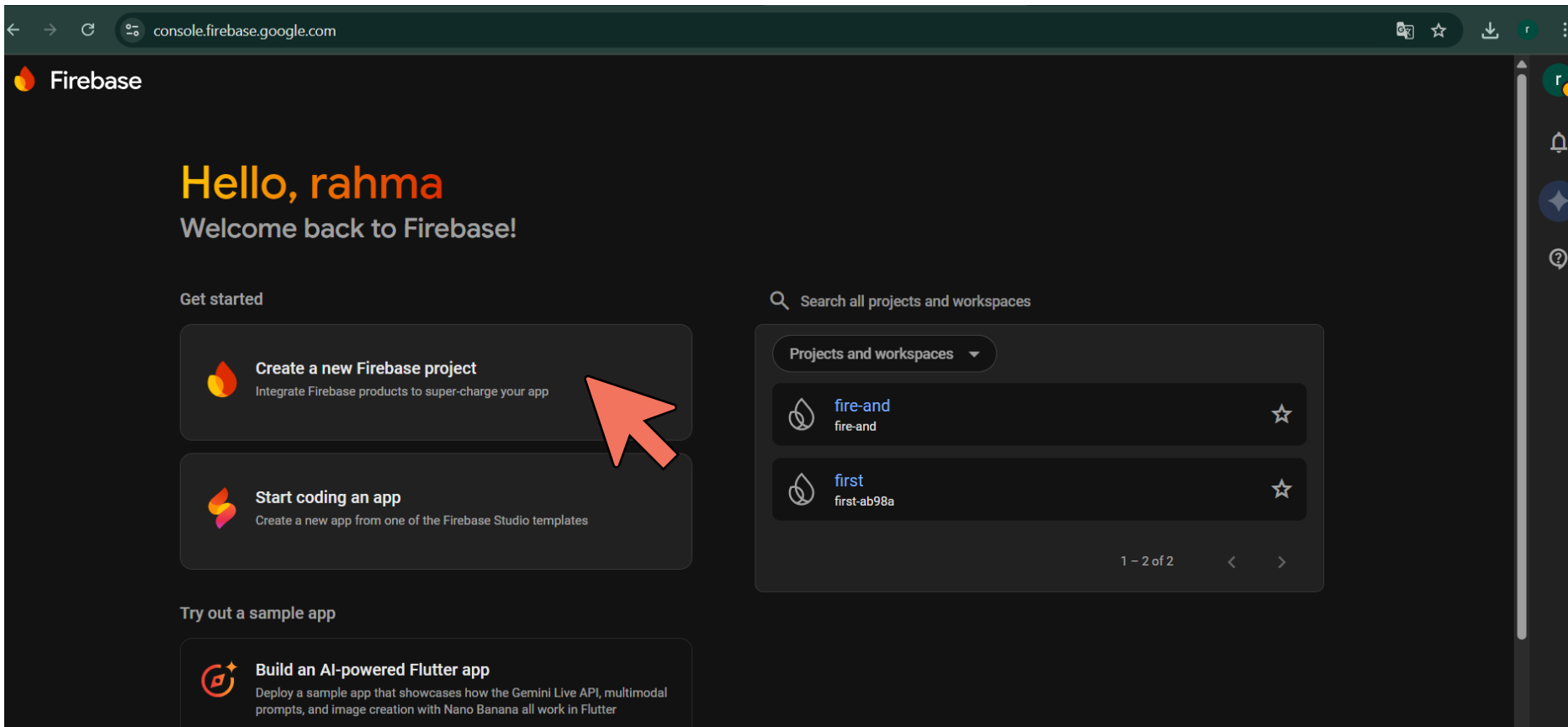
Prototype, build, and deploy full-stack, AI apps with our agentic development environment. Now with AI-optimized templates, seamless integration with Firebase services, and ability to fork workspaces, all designed to make AI-assisted development more powerful.

Get Started

Make your app 🚀 the best it can be
with Firebase and generative AI

Firebase: Creating a New Project

❖ Click on **create a new firebase project**.



The screenshot shows the Firebase console interface. At the top, the browser address bar displays 'console.firebase.google.com'. The main header includes the Firebase logo and the text 'Hello, rahma' and 'Welcome back to Firebase!'. Below this, the 'Get started' section contains two prominent buttons: 'Create a new Firebase project' (with a red arrow pointing to it) and 'Start coding an app'. To the right, a search bar is labeled 'Search all projects and workspaces', and below it, a list of existing projects is shown, including 'fire-and' and 'first'. The bottom section, 'Try out a sample app', features a button for 'Build an AI-powered Flutter app'.

console.firebase.google.com

Firebase

Hello, rahma

Welcome back to Firebase!

Get started

 **Create a new Firebase project**
Integrate Firebase products to super-charge your app

 **Start coding an app**
Create a new app from one of the Firebase Studio templates

Try out a sample app

 **Build an AI-powered Flutter app**
Deploy a sample app that showcases how the Gemini Live API, multimodal prompts, and image creation with Nano Banana all work in Flutter

Search all projects and workspaces

Projects and workspaces ▾

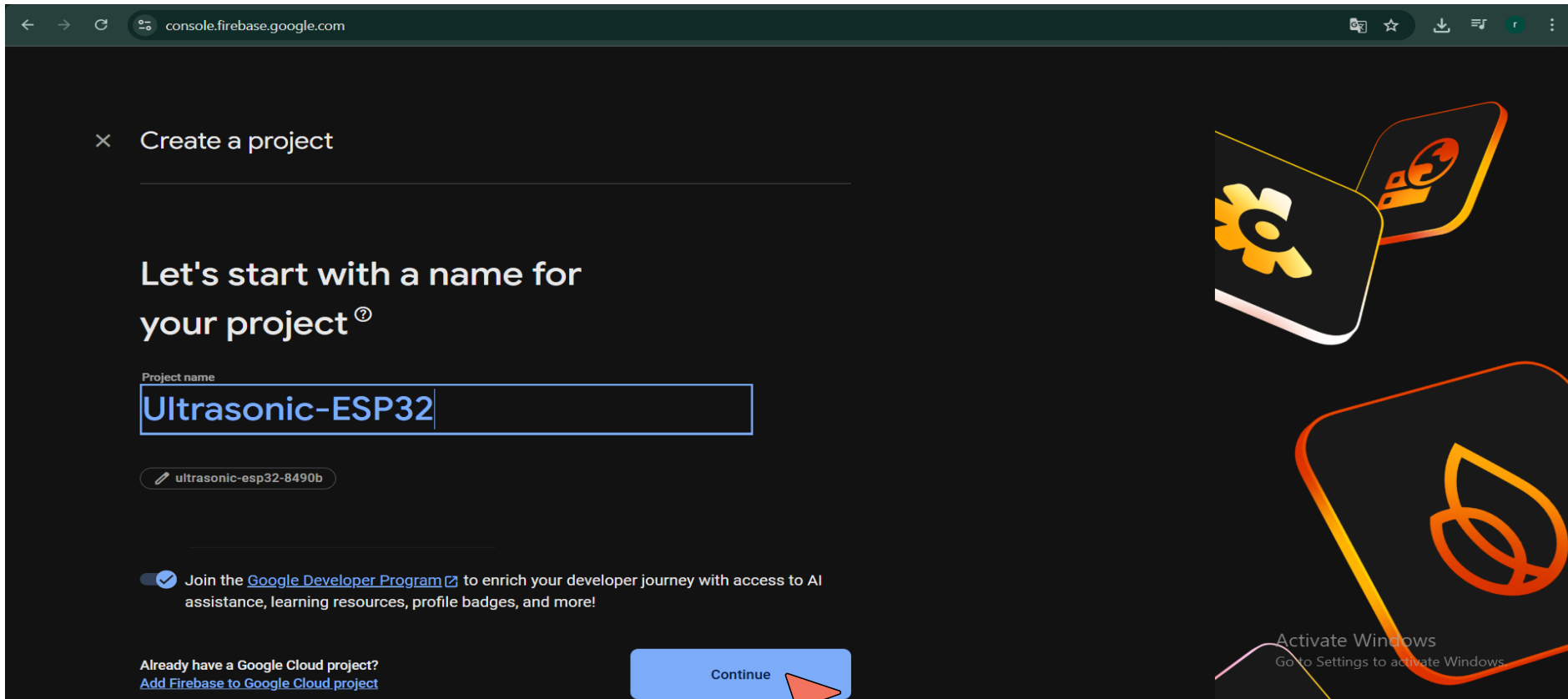
 **fire-and**
fire-and

 **first**
first-ab98a

1 – 2 of 2 < >

Firebase: Creating a New Project

❖ Write your **project name** and then continue.



× Create a project

Let's start with a name for your project[®]

Project name

Ultrasonic-ESP32

ultrasonic-esp32-8490b

☒ Join the [Google Developer Program](#) to enrich your developer journey with access to AI assistance, learning resources, profile badges, and more!

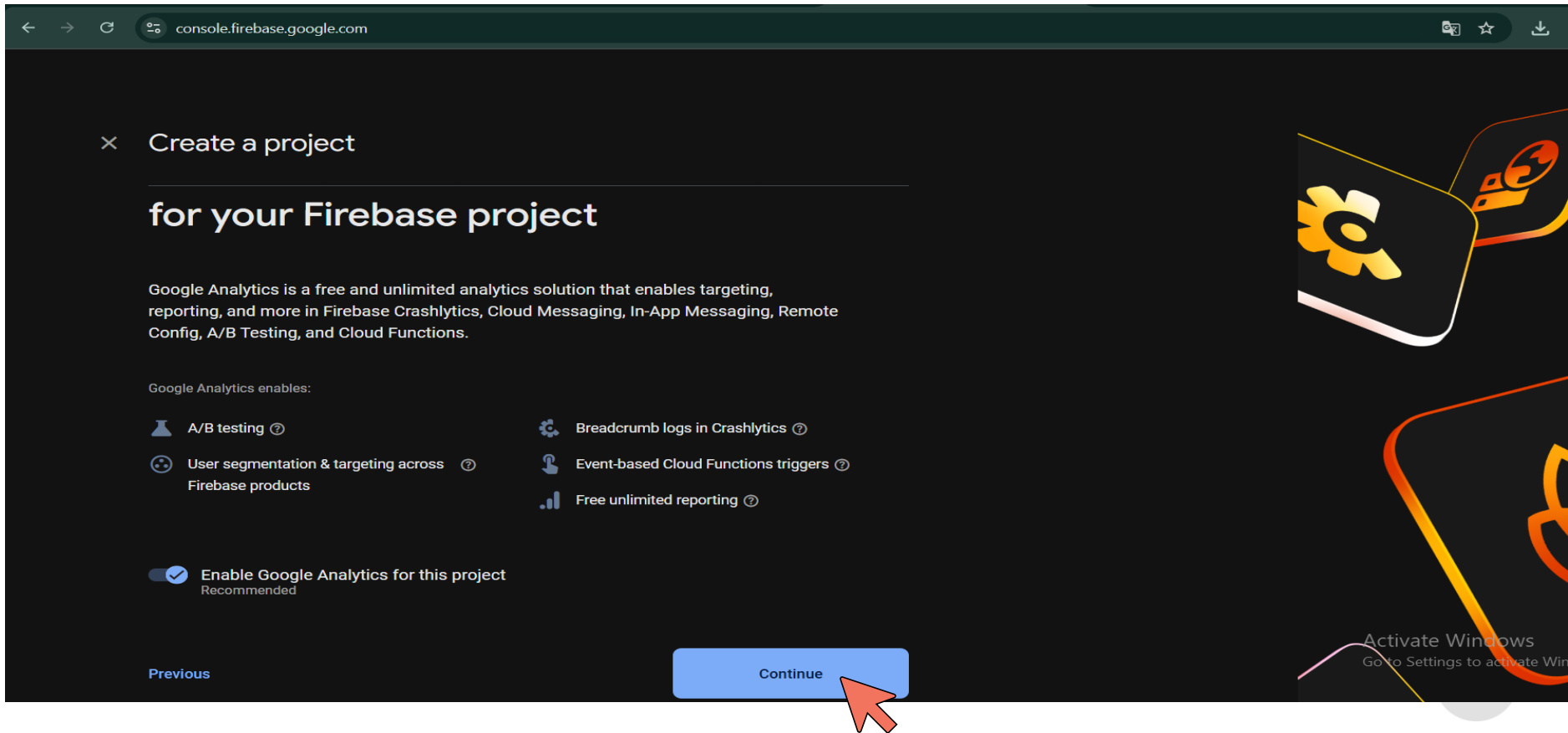
Already have a Google Cloud project?
[Add Firebase to Google Cloud project](#)

Continue

Activate Windows
Go to Settings to activate Windows

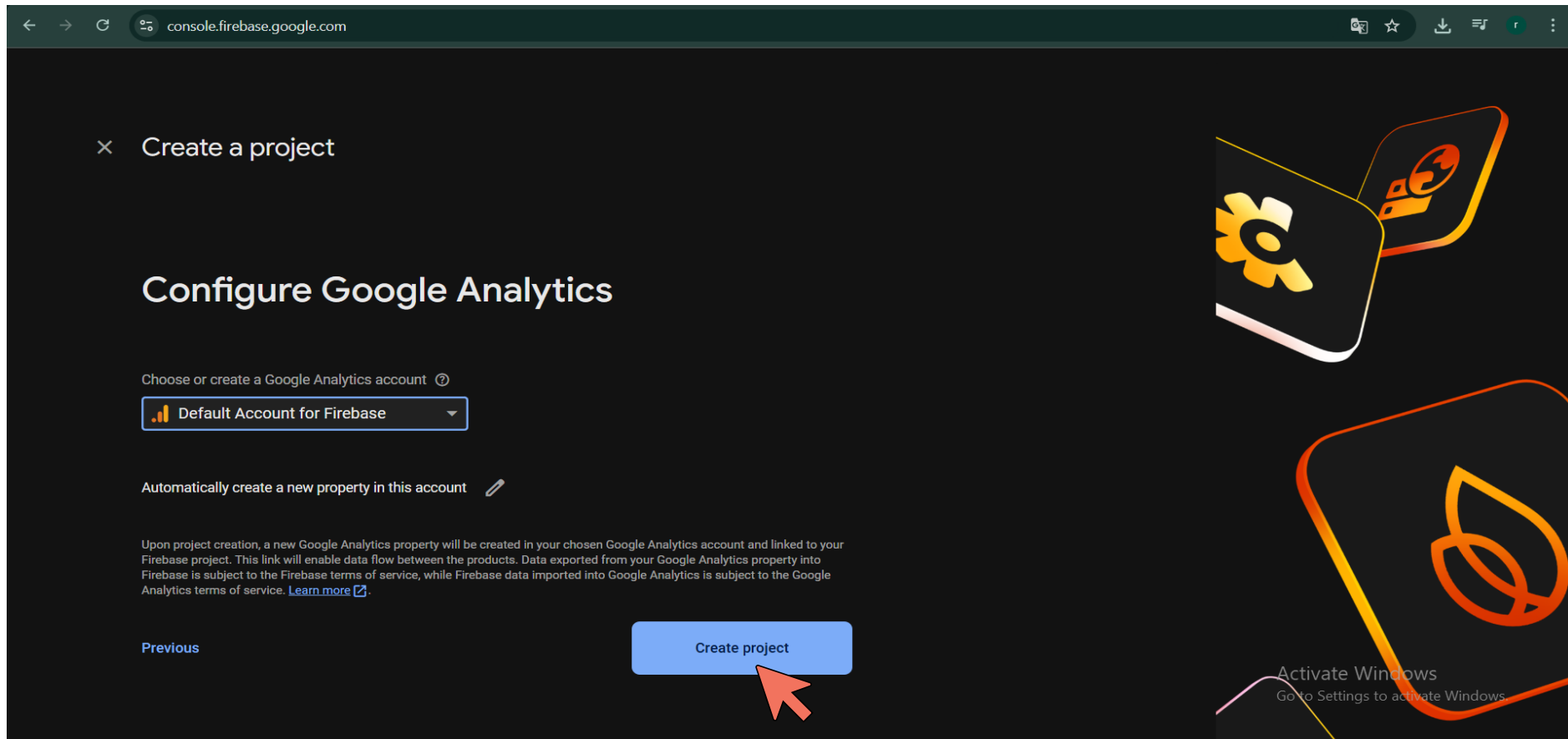
Firebase: Creating a New Project

❖ Click **continue**.



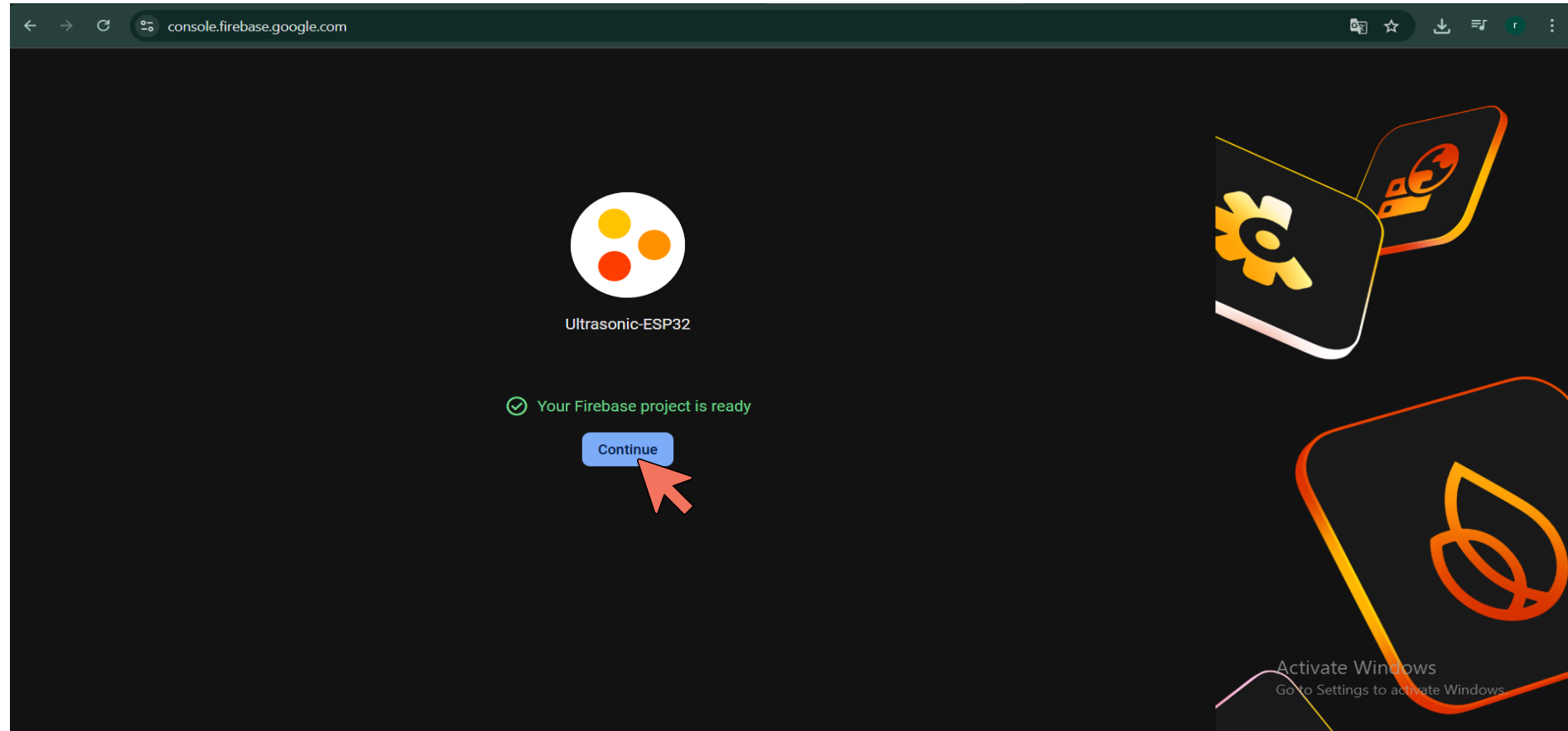
Firebase: Creating a New Project

- ❖ Choose **Default Account for firebase** and then click on **create project**.



Firebase: Creating a New Project

❖ Wait until project is created ,then click **continue**.



Firebase: Creating a New Project

❖ You will be redirected to your project consol page.

The screenshot shows the Firebase console interface. On the left is a dark sidebar with the 'Firebase' logo at the top. Below it is a 'Project Overview' button with a home icon and a settings gear icon. Underneath are sections for 'What's new' (highlighting 'AI Logic' with a 'NEW' badge), 'Product categories' (listing 'Build', 'Run', 'Analytics', and 'AI' with dropdown arrows), 'Related development tools' (featuring 'Firebase Studio' with an external link icon), and 'Spark' (labeled 'No-cost (\$0/month)' with an 'Upgrade' button and 'NEW' badge). The main content area has a dark background. At the top, it shows the project name 'Ultrasonic-ESP32' with a dropdown arrow. Below this is a banner for 'Try an agentic barista app powered by Firebase AI Logic!' with a 'Try Sample App' button and a close icon. The project name 'Ultrasonic-ESP32' is displayed again with a 'Spark plan' badge, followed by a '+ Add app' button. A large 'Hello, rahma' greeting is shown in orange and white, with 'Welcome to your Firebase project!' below it. A message states 'Gemini can give you suggestions for Firebase services and solutions for your app'. A 'Next steps with Gemini' section contains four cards: 'Tell us about your app' (highlighted with a yellow border), 'Add Analytics and monitoring', 'Build a backend', and 'Add AI to your app'. Each card has a blue star icon and a brief description. A 'View docs' link is to the right of the first card. At the bottom right, an 'Activate Windows' watermark is visible.

Firebase

Ultrasonic-ESP32 ▾

[Project Overview](#)

What's new

AI Logic **NEW**

Product categories

- Build ▾
- Run ▾
- Analytics ▾
- AI ▾

Related development tools

[Firebase Studio](#)

Spark No-cost (\$0/month) **Upgrade** **NEW**

Try an agentic barista app powered by Firebase AI Logic! [Try Sample App](#)

Ultrasonic-ESP32 **Spark plan**

[+ Add app](#)

Hello, rahma

Welcome to your Firebase project!

Gemini can give you suggestions for Firebase services and solutions for your app

Next steps with Gemini [View docs](#)

- Tell us about your app**
Describe your app, and Gemini will suggest products to help you get started
- Add Analytics and monitoring**
Learn more about Google Analytics and Firebase monitoring products
- Build a backend**
Firebase offers many backend services, including both SQL and NoSQL database options. Learn more about which services might
- Add AI to your app**
Learn more about integrating AI into your app

Activate Windows
Go to Settings to activate Windows.

Firebase: Creating a New Project

❖ Expand **Build** tap and go to **Authentication**.

The screenshot shows the Firebase console interface for a project named 'Ultrasonic-ESP32'. On the left sidebar, the 'Build' menu is expanded, and 'Authentication' is highlighted with a red arrow. The main content area shows the project overview with a 'Spark plan' badge and an 'Add app' button. Below this, the 'Analytics' section is visible, showing 'Daily active users' and 'Day 1 retention' metrics, both of which are currently 'No data for the last 14 days'. A 'Track your revenue!' section with links to 'AdMob' and 'Google Play' is also present. The bottom of the screen shows an 'Activate Windows' watermark.

Ultrasonic-ESP32

Try an agentic barista app powered by Firebase AI Logic! Try Sample App

Ultrasonic-ESP32 Spark plan

esp32app + Add app

Analytics

Analytics

Daily active users Day 1 retention

No data for the last 14 days No data for the last 14 days

Track your revenue!

Link to AdMob Link to Google Play

Activate Windows

Go to Settings to activate Windows.

This week Last week

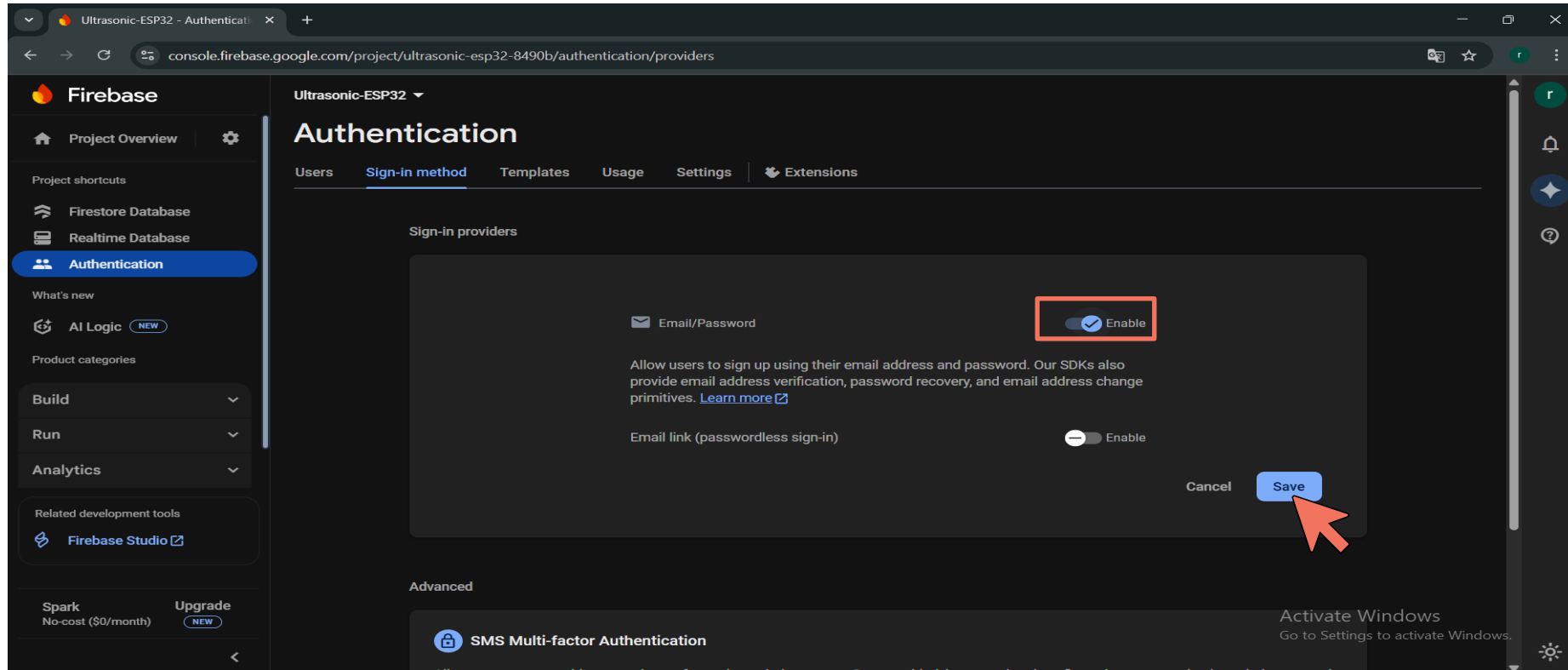
Firebase: Creating a New Project

❖ Go to **Sign-in method** tap and click on **Email/password**.

The screenshot shows the Firebase Authentication console for the project 'Ultrasonic-ESP32'. The left sidebar contains navigation links: Project Overview, Project shortcuts (Firestore Database, Realtime Database), Authentication (selected), What's new, AI Logic (NEW), Product categories (Build, Run, Analytics), and Related development tools (Firebase Studio). The main content area is titled 'Authentication' and has tabs for Users, Sign-in method (highlighted with a red box), Templates, Usage, Settings, and Extensions. Under the 'Sign-in method' tab, there's a section 'Sign-in providers' with the heading 'Get started with Firebase Auth by adding your first sign-in method'. This section is divided into three columns: Native providers (Email/Password, Phone, Anonymous), Additional providers (Google, Facebook, Play Games, Game Center, Apple, GitHub, Microsoft, Twitter, Yahoo), and Custom providers (OpenID Connect, SAML). A red arrow points to the 'Email/Password' option in the Native providers column. Below this, there's an 'Advanced' section with 'SMS Multi-factor Authentication' and a note about adding an extra layer of security. At the bottom right, there's a 'Activate Windows' watermark.

Firebase: Creating a New Project

❖ Click on **Enable** then click on **save**.



Firebase: Creating a New Project

❖ Now you have created a provider, after that **click on Users**.

The screenshot shows the Firebase console interface for the project 'Ultrasonic-ESP32'. The left sidebar contains navigation links for Project Overview, Project shortcuts (Firestore Database, Realtime Database), Authentication (highlighted), What's new, AI Logic (NEW), Product categories, Build, Run, Analytics, Related development tools (Firebase Studio), and Spark (No-cost (\$0/month) with an Upgrade button). The main content area is titled 'Authentication' and has tabs for Users, Sign-in method (selected), Templates, Usage, Settings, and Extensions. Under 'Sign-in providers', there is a table with one provider: 'Email/Password' with a status of 'Enabled'. An 'Add new provider' button is in the top right. The 'Advanced' section features 'SMS Multi-factor Authentication' with a description and a 'Learn more' link. A star icon indicates that MFA and other advanced features are available with Identity Platform. At the bottom right, there is a banner for 'Upgrade to enable Windows' with a link to 'Go to Settings to activate Windows'.

Provider	Status
Email/Password	Enabled

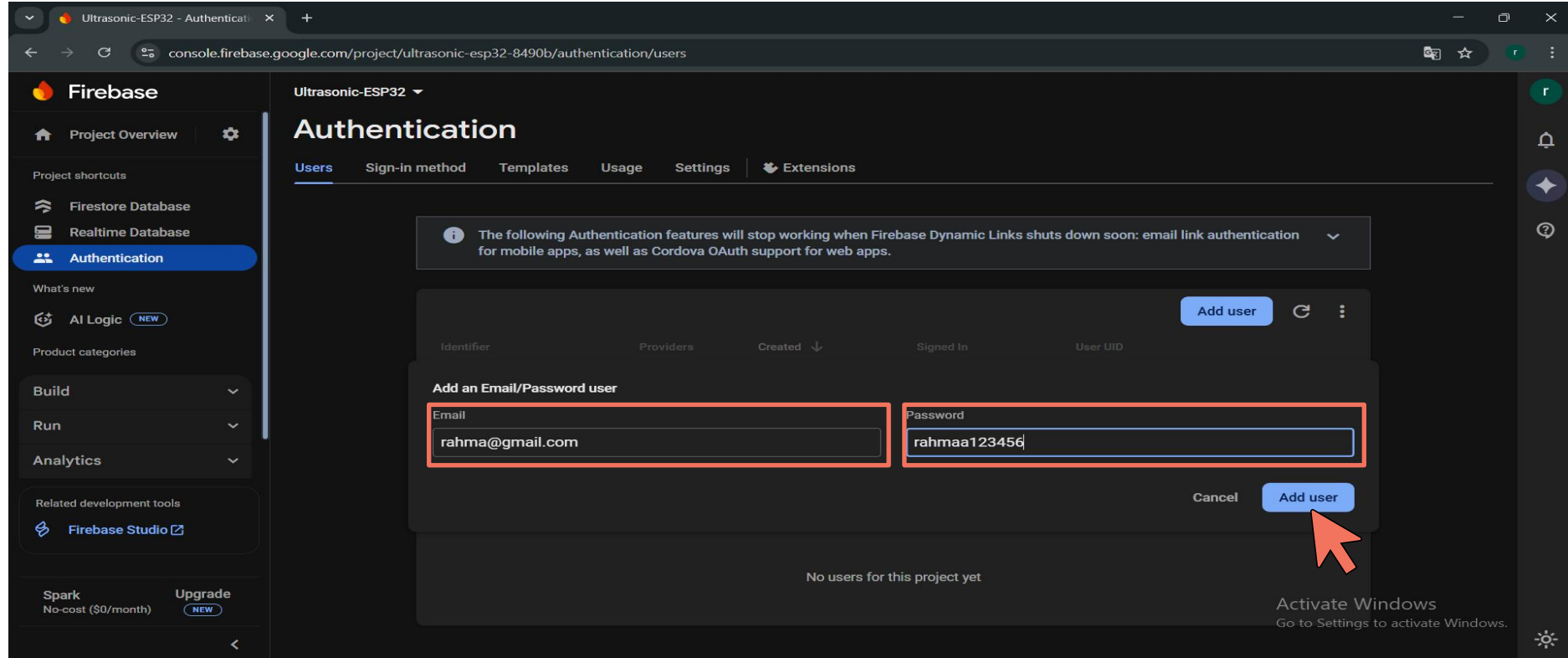
Firebase: Creating a New Project

❖ click on Add User.

The screenshot shows the Firebase Authentication console for the project 'Ultrasonic-ESP32'. The left sidebar contains navigation links for Project Overview, Firestore Database, Realtime Database, Authentication (selected), AI Logic, and Analytics. The main content area is titled 'Authentication' and includes tabs for Users, Sign-in method, Templates, Usage, Settings, and Extensions. A warning message at the top states: 'The following Authentication features will stop working when Firebase Dynamic Links shuts down soon: email link authentication for mobile apps, as well as Cordova OAuth support for web apps.' Below this, there is a table with columns: Identifier, Providers, Created, Signed In, and User UID. The table is currently empty, displaying 'No users for this project yet'. A blue 'Add user' button is located in the top right corner of the table area, and a red arrow points to it. The bottom right corner of the screen shows a Windows watermark: 'Activate Windows Go to Settings to activate Windows.'

Firebase: Creating a New Project

❖ Write a **user email and password** then click on **Add User**.



The screenshot shows the Firebase Authentication console for the project 'Ultrasonic-ESP32'. The 'Users' tab is selected. A modal dialog titled 'Add an Email/Password user' is open, with the 'Email' field containing 'rahma@gmail.com' and the 'Password' field containing 'rahmaa123456'. Both fields are highlighted with red boxes. A red arrow points to the 'Add user' button in the bottom right corner of the dialog. The background shows the Firebase console interface with a sidebar on the left containing navigation links like 'Project Overview', 'Firestore Database', 'Realtime Database', and 'Authentication'. A table with columns 'Identifier', 'Providers', 'Created', 'Signed in', and 'User UID' is visible below the dialog, showing 'No users for this project yet'.

Ultrasonic-ESP32

Authentication

Users | Sign-in method | Templates | Usage | Settings | Extensions

The following Authentication features will stop working when Firebase Dynamic Links shuts down soon: email link authentication for mobile apps, as well as Cordova OAuth support for web apps.

Add user

Identifier	Providers	Created	Signed in	User UID
Add an Email/Password user				
Email				
rahma@gmail.com				
Password				
rahmaa123456				

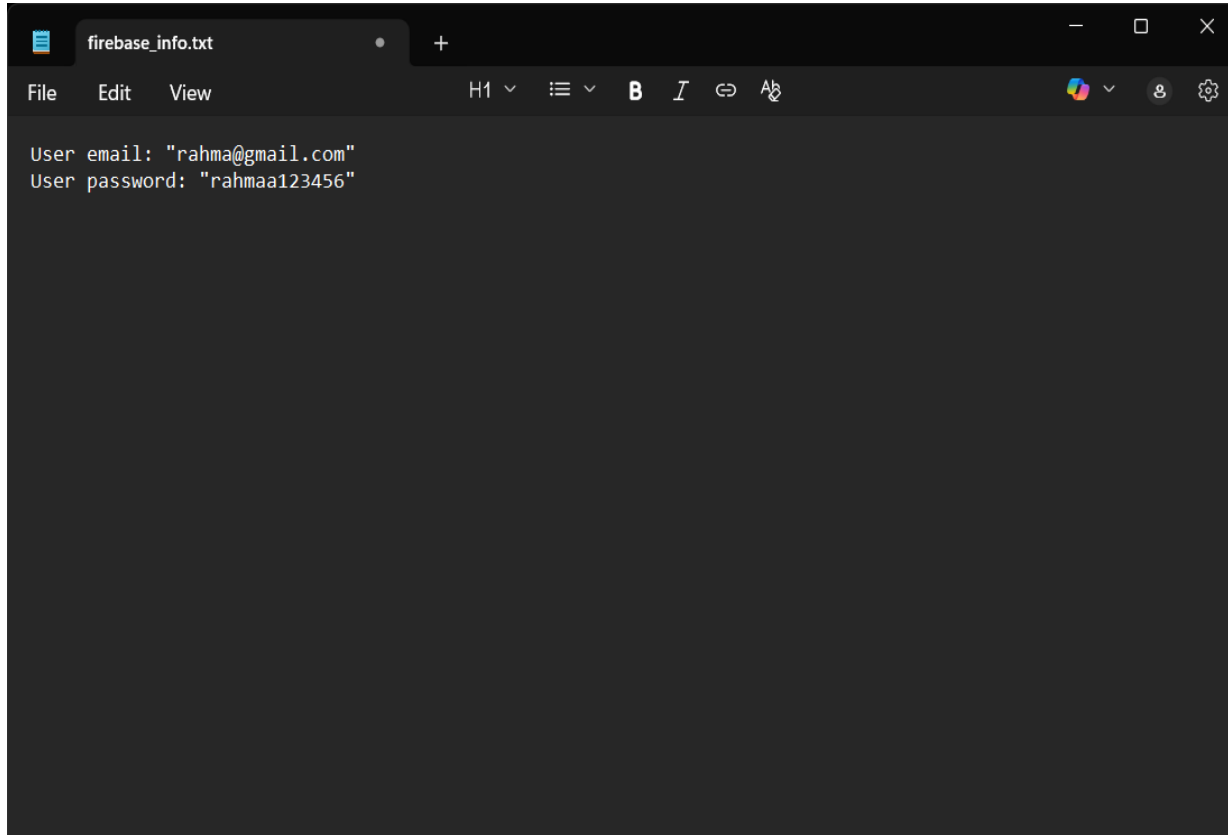
Cancel Add user

No users for this project yet

Activate Windows
Go to Settings to activate Windows.

Firebase: Creating a Realtime Database

❖ Save your user email and password in file to use them later in the ESP32 code.

A screenshot of a text editor window with a dark theme. The title bar shows a single tab named 'firebase_info.txt'. The menu bar includes 'File', 'Edit', and 'View'. The toolbar contains icons for font size (H1), list view, bold (B), italic (I), link, and undo. The main text area contains two lines of code: 'User email: "rahma@gmail.com"' and 'User password: "rahmaa123456"'.

```
User email: "rahma@gmail.com"
User password: "rahmaa123456"
```



Firebase: Creating a Realtime Database

❖ Then, Expand **Build** tap and go to **Realtime Database**.

The screenshot shows the Firebase console interface. On the left, the 'Build' menu is expanded, showing various services. A red arrow points to the 'Realtime Database' option. The main area displays the 'Ultrasonic-ESP32' project with a 'Hello, rahma' greeting and a 'Welcome to your Firebase project!' message. Below this, there are 'Next steps with Gemini' suggestions, including 'Tell us about your app', 'Add Analytics and monitoring', 'Build a backend', and 'Add AI to your app'. The bottom of the screen shows the 'Spark' plan and an 'Upgrade' button.

Firebase

Realtime Database

What's new

AI Logic NEW

Product categories

Build

- App Check
- App Hosting
- Authentication
- Data Connect
- Extensions
- Firestore Database
- Functions
- Hosting
- Machine Learning
- Realtime Database**
- Storage

Spark No-cost (\$0/month) Upgrade NEW

Ultrasonic-ESP32

Try an agentic barista app powered by Firebase AI Logic! Try Sample App

Ultrasonic-ESP32 Spark plan

+ Add app

Hello, rahma

Welcome to your Firebase project!

Gemini can give you suggestions for Firebase services and solutions for your app

Next steps with Gemini [View docs](#)

- Tell us about your app**
Describe your app, and Gemini will suggest products to help you get started
- Add Analytics and monitoring**
Learn more about Google Analytics and Firebase monitoring products
- Build a backend**
Firebase offers many backend services, including both SQL and NoSQL database options. Learn more about which services might
- Add AI to your app**
Learn more about integrating AI into your app

Activate Windows
Go to Settings to activate Windows.

<https://console.firebase.google.com/project/ultrasonic-esp32-8490b/database>

Firebase: Creating a Realtime Database

❖ Click on **Create Database**.

The screenshot displays the Firebase console interface. On the left, a sidebar contains the following elements: the Firebase logo, 'Project Overview' with a settings icon, 'Project shortcuts' with links to 'Firestore Database' and 'Realtime Database' (the latter is highlighted in blue), 'What's new' with an 'AI Logic' badge labeled 'NEW', 'Product categories' with expandable sections for 'Build', 'Run', 'Analytics', and 'AI', 'Related development tools' with a link to 'Firebase Studio', and 'Spark' (No-cost \$0/month) and 'Upgrade' (NEW) buttons. The main content area is titled 'Realtime Database' with the subtitle 'Store and sync data in real time'. It features two buttons: 'Create Database' (highlighted with a red arrow) and 'Ask Gemini'. To the right is an illustration of server racks with a checkmark and a network diagram. At the bottom, there is a 'Learn more' section and a banner for 'Introducing Firebase Realtime Database' with a red arrow icon. The Windows taskbar is visible at the very bottom, showing the system clock and an 'Activate Windows' watermark.

Ultrasonic-ESP32 ▾

Realtime Database

Store and sync data in real time

[Create Database](#) [Ask Gemini](#)

Is Realtime Database right for you? [Compare Databases](#)

Learn more

Introducing Firebase Realtime Database

Activate Windows
Go to Settings to activate Windows.

Firebase: Creating a Realtime Database

❖ Select **United States** as your **database location** then and click **Next**.

The screenshot shows the Firebase console interface. On the left, the sidebar contains the following sections:

- Project Overview** (with a settings gear icon)
- Project shortcuts**
 - Firestore Database
 - Realtime Database** (highlighted in blue)
- What's new**
 - AI Logic (with a 'NEW' badge)
- Product categories**
 - Build
 - Run
 - Analytics
 - AI
- Related development tools**
 - Firebase Studio
- Spark** (No-cost (\$0/month)) and **Upgrade** (with a 'NEW' badge)

The main content area displays the 'Set up database' dialog for the project 'Ultrasonic-ESP32'. The dialog has two tabs: '1 Database options' (active) and '2 Security rules'. The text states: 'Your location setting is where your Realtime Database data will be stored.' Below this, the 'Realtime Database location' dropdown menu is open, showing the following options:

- United States (us-central1) (selected)
- United States (us-central1)
- Belgium (europe-west1)
- Singapore (asia-southeast1)

At the bottom right of the dialog, there are 'Cancel' and 'Next' buttons. A red arrow points to the 'Next' button. In the background, there is a faint diagram of a database structure with nodes and connections.

At the bottom of the screen, there is a banner for 'Introducing Firebase Realtime Database' and a Windows watermark that says 'Activate Windows Go to Settings to activate Windows.'

Firebase: Creating a Realtime Database

❖ Select **Start in test mode** then click **enable**. I chose *start in test mode* to easily read and write data during testing without authentication

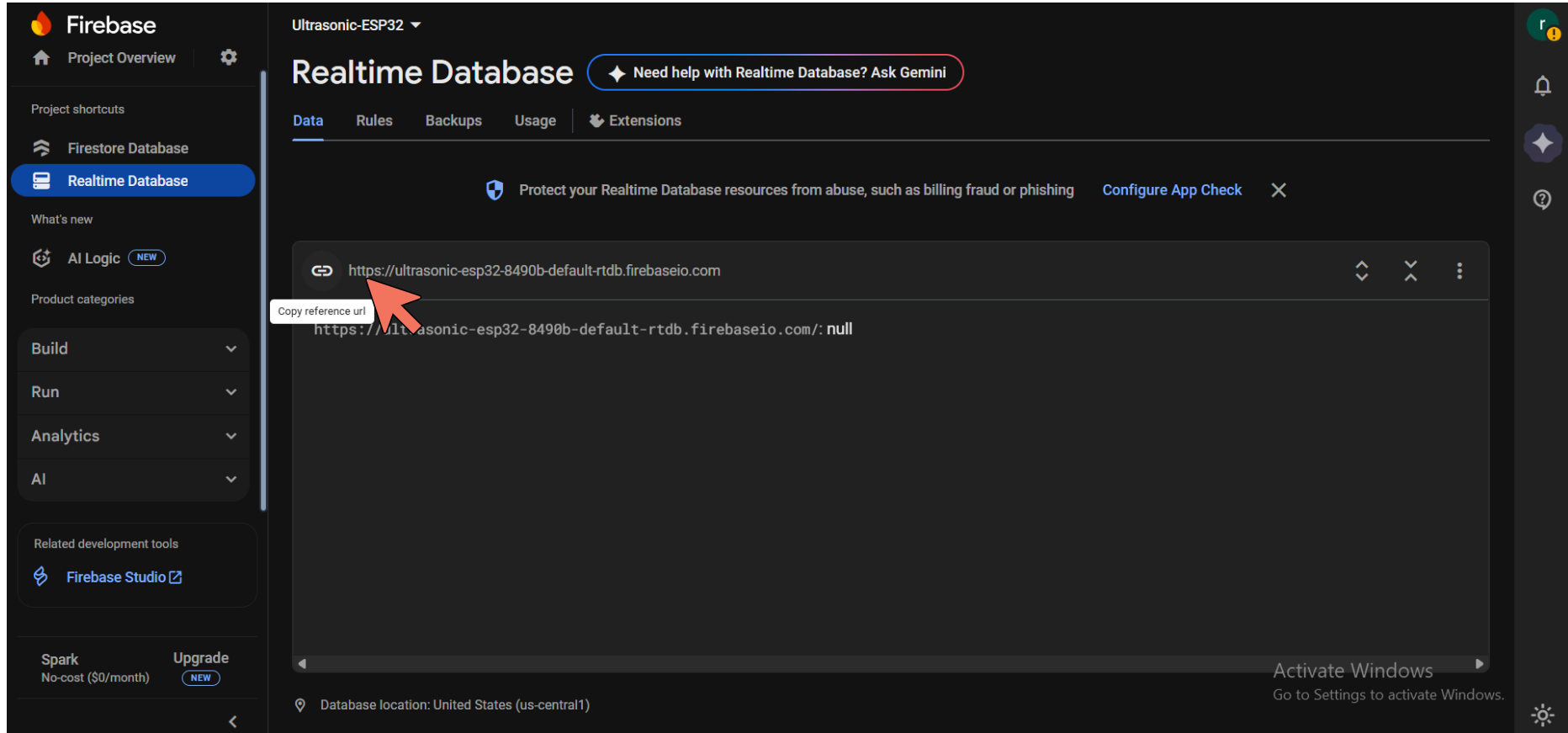
The screenshot shows the Firebase console interface for a project named 'Ultrasonic-ESP32'. The left sidebar contains navigation links for Project Overview, Project shortcuts (Firestore Database, Realtime Database), What's new (AI Logic), and Product categories (Build, Run, Analytics, AI). The main area displays the 'Set up database' dialog with two tabs: 'Database options' and 'Security rules'. The 'Security rules' tab is active, showing two options: 'Start in locked mode' and 'Start in test mode'. The 'Start in test mode' option is selected, indicated by a blue circle. A warning box with a yellow exclamation mark states: 'The default security rules for test mode allow anyone with your database reference to view, edit and delete all data in your database for the next 30 days'. Below the warning, the 'Enable' button is highlighted with a red arrow. The dialog also includes a 'Cancel' button and a code block showing the default security rules for test mode.

```
{
  "rules": {
    ".read": "now < 1763330400000", // 2025-11-17
    ".write": "now < 1763330400000", // 2025-11-17
  }
}
```

At the bottom of the screen, there is a banner for 'Introducing Firebase Realtime Database' and a Windows activation watermark.

Firebase: Creating a Realtime Database

❖ Now your Database is created, you need to copy your database URL to use it in the ESP32 code for connecting to firebase.



The screenshot shows the Firebase console interface for a project named "Ultrasonic-ESP32". The left sidebar contains navigation options: "Project Overview", "Firestore Database", "Realtime Database" (highlighted), "AI Logic", and "Product categories". The main area displays the "Realtime Database" settings, including tabs for "Data", "Rules", "Backups", "Usage", and "Extensions". A notification bar states: "Protect your Realtime Database resources from abuse, such as billing fraud or phishing". Below this, a code editor shows the database URL: `https://ultrasonic-esp32-8490b-default-rtdb.firebaseio.com`. A red arrow points to this URL, and a tooltip "Copy reference url" is visible. The bottom of the console shows the "Database location: United States (us-central1)".

Ultrasonic-ESP32 ▾

Realtime Database

◆ Need help with Realtime Database? Ask Gemini

Data Rules Backups Usage Extensions

Protect your Realtime Database resources from abuse, such as billing fraud or phishing [Configure App Check](#) ✕

```
https://ultrasonic-esp32-8490b-default-rtdb.firebaseio.com
```

Copy reference url

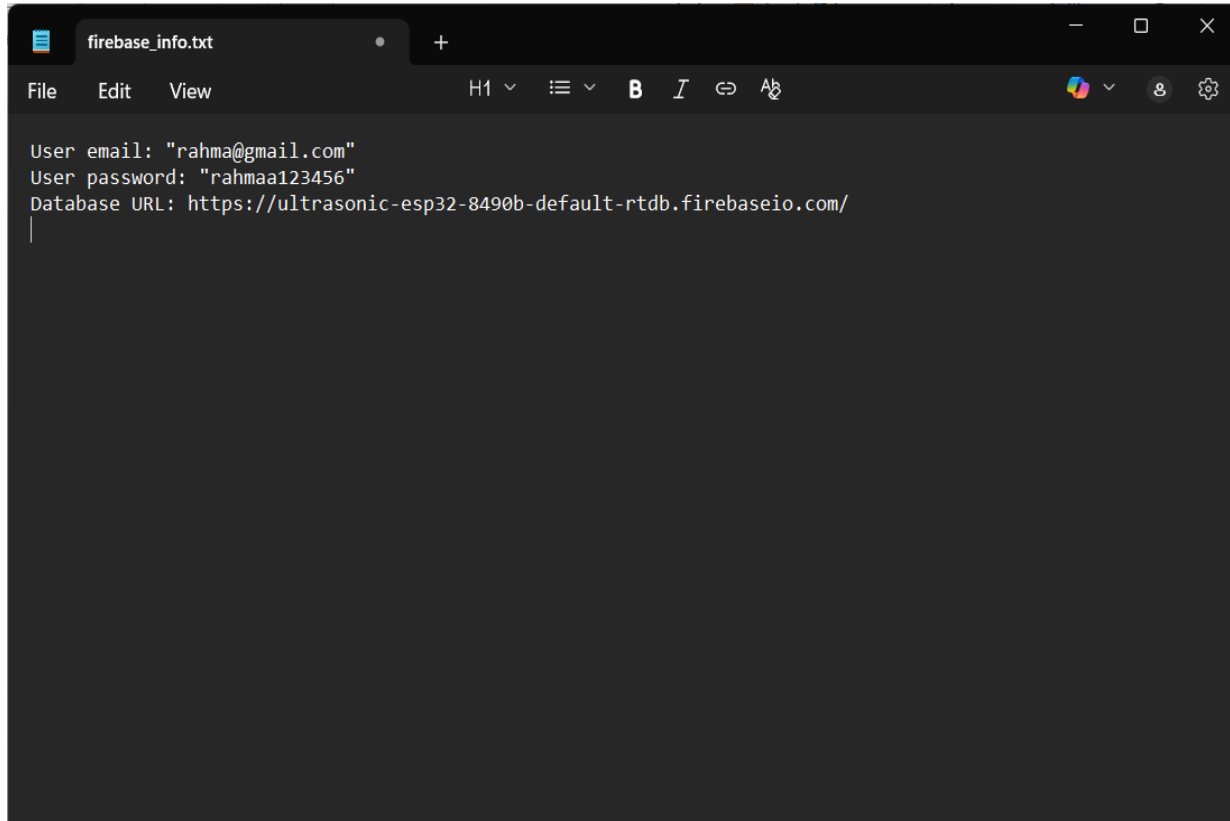
```
https://ultrasonic-esp32-8490b-default-rtdb.firebaseio.com/: null
```

Database location: United States (us-central1)

Activate Windows
Go to Settings to activate Windows.

Firebase: Creating a Realtime Database

❖ Update your Notepad file with your Database URL to use them later in the ESP32 code.



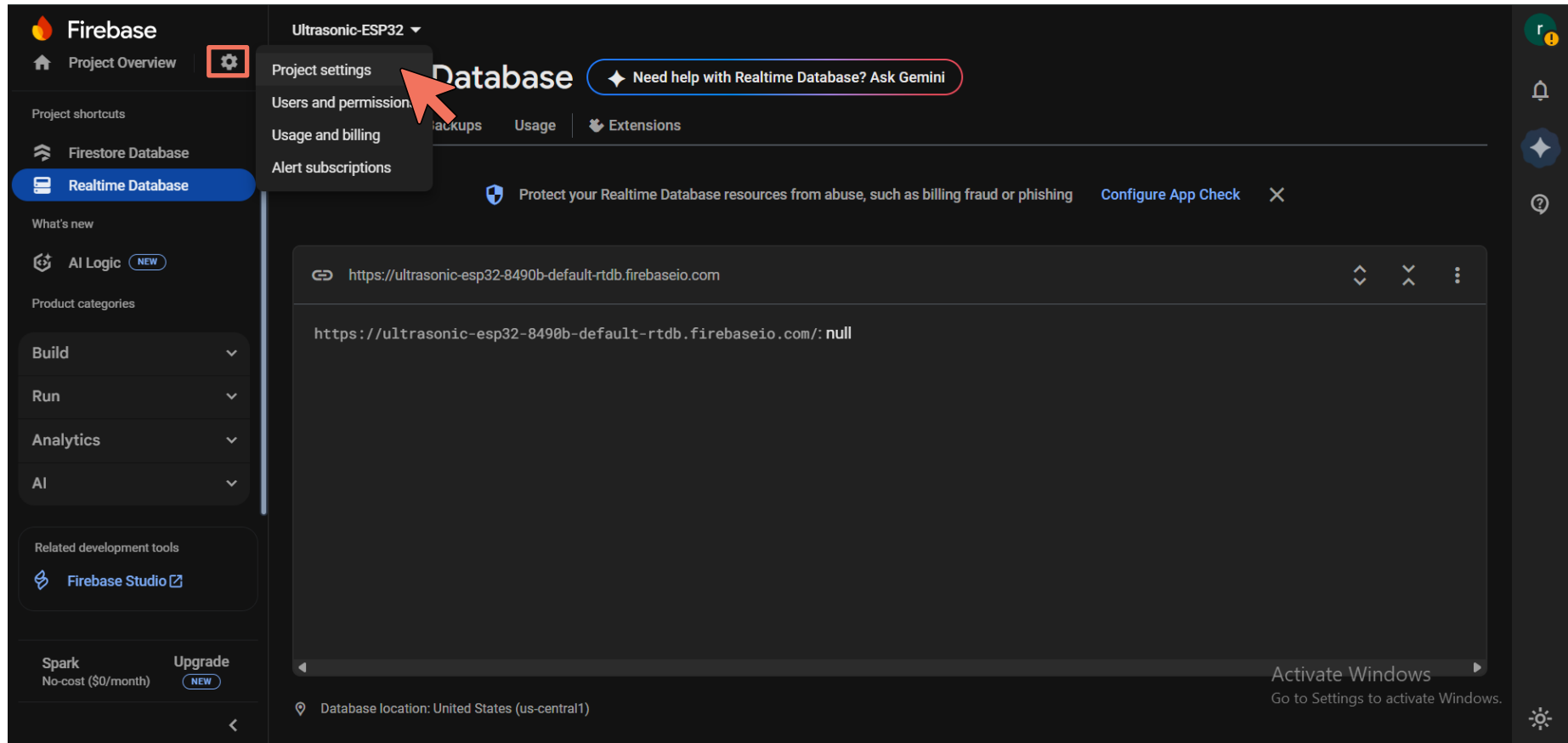
A screenshot of a Notepad application window. The title bar shows 'firebase_info.txt'. The menu bar includes 'File', 'Edit', and 'View'. The toolbar shows 'H1', a list icon, 'B' (bold), 'I' (italic), '↺' (undo), and '↻' (redo). The text area contains the following text:

```
User_email: "rahma@gmail.com"  
User_password: "rahmaa123456"  
Database_URL: https://ultrasonic-esp32-8490b-default-rtdb.firebaseio.com/  
|
```



Firebase: Creating a Realtime Database

❖ Go to **project settings**.



The screenshot shows the Firebase console interface. On the left sidebar, the 'Project settings' option is highlighted with a red box and a red arrow. The main content area displays the 'Realtime Database' settings for the project 'Ultrasonic-ESP32'. The URL for the database is shown as `https://ultrasonic-esp32-8490b-default-rtdb.firebaseio.com`. Below the URL, the database location is specified as 'United States (us-central1)'. A 'Need help with Realtime Database? Ask Gemini' button is visible at the top right of the main content area. The bottom right corner of the screen shows a Windows watermark: 'Activate Windows Go to Settings to activate Windows.'

Firebase: Creating a Realtime Database

❖ Scroll down until you reach **your apps window**.

The screenshot shows the Firebase console interface. On the left is a dark sidebar with navigation options: 'Project Overview' (selected), 'Project shortcuts' (Firestore Database, Realtime Database), 'What's new' (AI Logic), and 'Product categories' (Build, Run, Analytics, AI). At the bottom of the sidebar are 'Related development tools' (Firebase Studio) and a 'Spark' promotion. The main content area is titled 'Project settings' for 'Ultrasonic-ESP32'. It has tabs for 'General', 'Cloud Messaging', 'Integrations', 'Service accounts', 'Data privacy', 'Users and permissions', and 'Alerts'. The 'General' tab is active, showing project details: Project name (Ultrasonic-ESP32), Project ID (ultrasonic-esp32-8490b), Project number (754507425265), and Web API Key (No Web API Key for this project). Below this is the 'Environment' section with the note 'This setting customizes your project for different stages of the app lifecycle' and 'Environment type' (Unspecified). At the bottom of the main area, a 'Your apps' button is highlighted with a red rectangle. On the far right, there is a vertical sidebar with a user profile, a bell icon, a star icon, and a help icon. At the very bottom right, there is an 'Activate Windows' watermark.

Project settings

General Cloud Messaging Integrations Service accounts Data privacy Users and permissions Alerts

Your project

Project name	Ultrasonic-ESP32
Project ID	ultrasonic-esp32-8490b
Project number	754507425265
Web API Key	No Web API Key for this project

Environment

This setting customizes your project for different stages of the app lifecycle

Environment type	Unspecified
------------------	-------------

Your apps

Activate Windows
Go to Settings to activate Windows.

Firebase: Creating a Realtime Database

❖ Select **web** app icon .

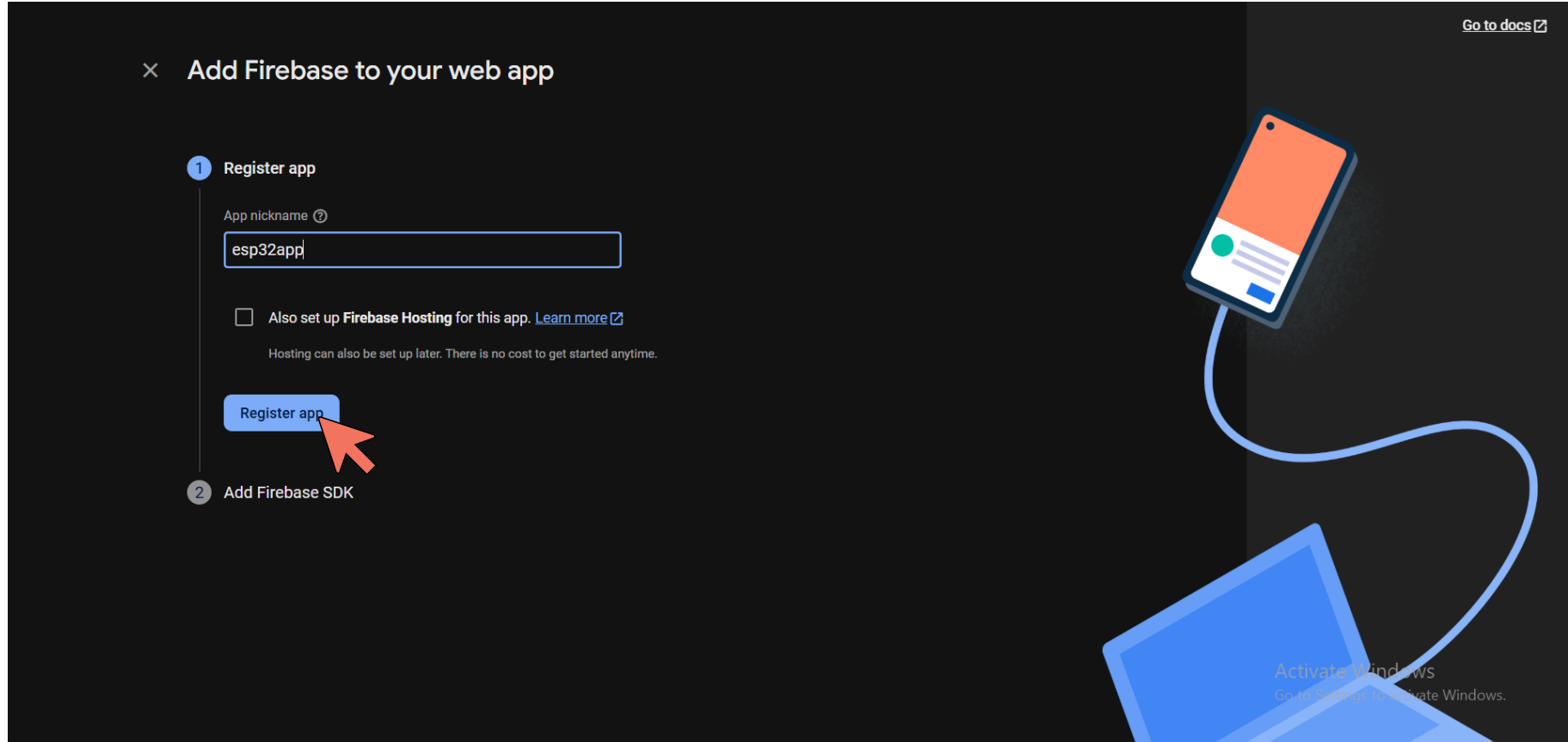
The screenshot shows the Firebase console interface. On the left, the sidebar contains the 'Project Overview' tab with a settings gear icon, 'Project shortcuts' for Firestore Database, Realtime Database, and Authentication, 'What's new' with 'AI Logic' (NEW), and 'Product categories' for Build, Run, and Analytics. At the bottom of the sidebar are 'Related development tools' with 'Firebase Studio' and 'Spark' (No-cost \$0/month) with an 'Upgrade' button (NEW).

The main content area is titled 'Project settings' for the project 'Ultrasonic-ESP32'. It displays project details: Project name (Ultrasonic-ESP32), Project ID (ultrasonic-esp32-8490b), Project number (754507425265), and Web API Key (No Web API Key for this project). Below this is the 'Environment' section, stating 'This setting customizes your project for different stages of the app lifecycle' and 'Environment type' (Unspecified).

The 'Your apps' section shows 'There are no apps in your project' and 'Select a platform to get started'. It features five circular icons: 'iOS+', Android, a web icon (highlighted with a red arrow), Apple TV, and Windows. An 'Activate Windows' watermark is visible in the bottom right corner.

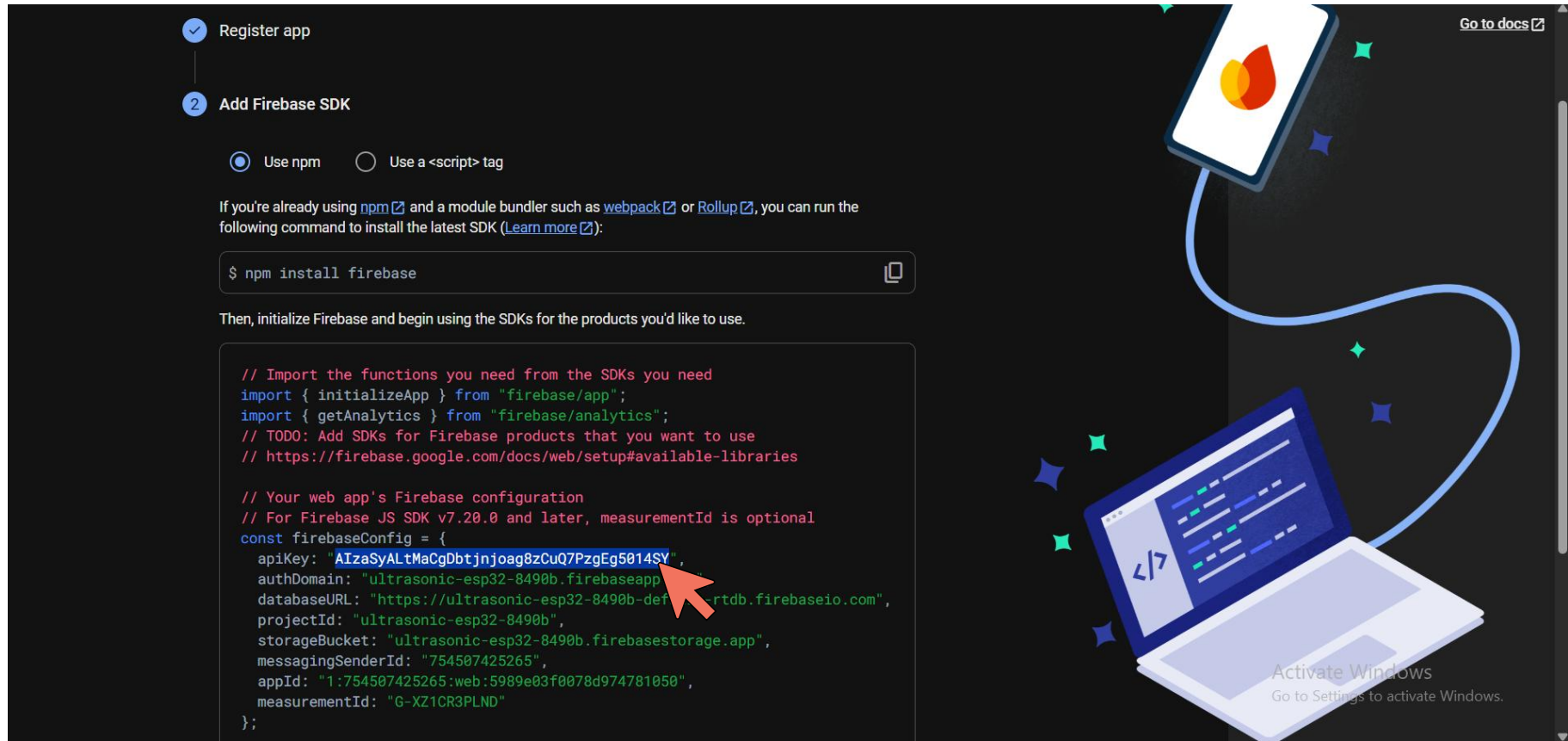
Firebase: Creating a Realtime Database

❖ Write your **App name** and then click **Register app**.



Firebase: Creating a Realtime Database

❖ Copy this **ApiKey**.



Register app

2 Add Firebase SDK

☒ Use npm ☐ Use a <script> tag

If you're already using [npm](#) and a module bundler such as [webpack](#) or [Rollup](#), you can run the following command to install the latest SDK ([Learn more](#)):

```
$ npm install firebase
```

Then, initialize Firebase and begin using the SDKs for the products you'd like to use.

```
// Import the functions you need from the SDKs you need
import { initializeApp } from "firebase/app";
import { getAnalytics } from "firebase/analytics";
// TODO: Add SDKs for Firebase products that you want to use
// https://firebase.google.com/docs/web/setup#available-libraries

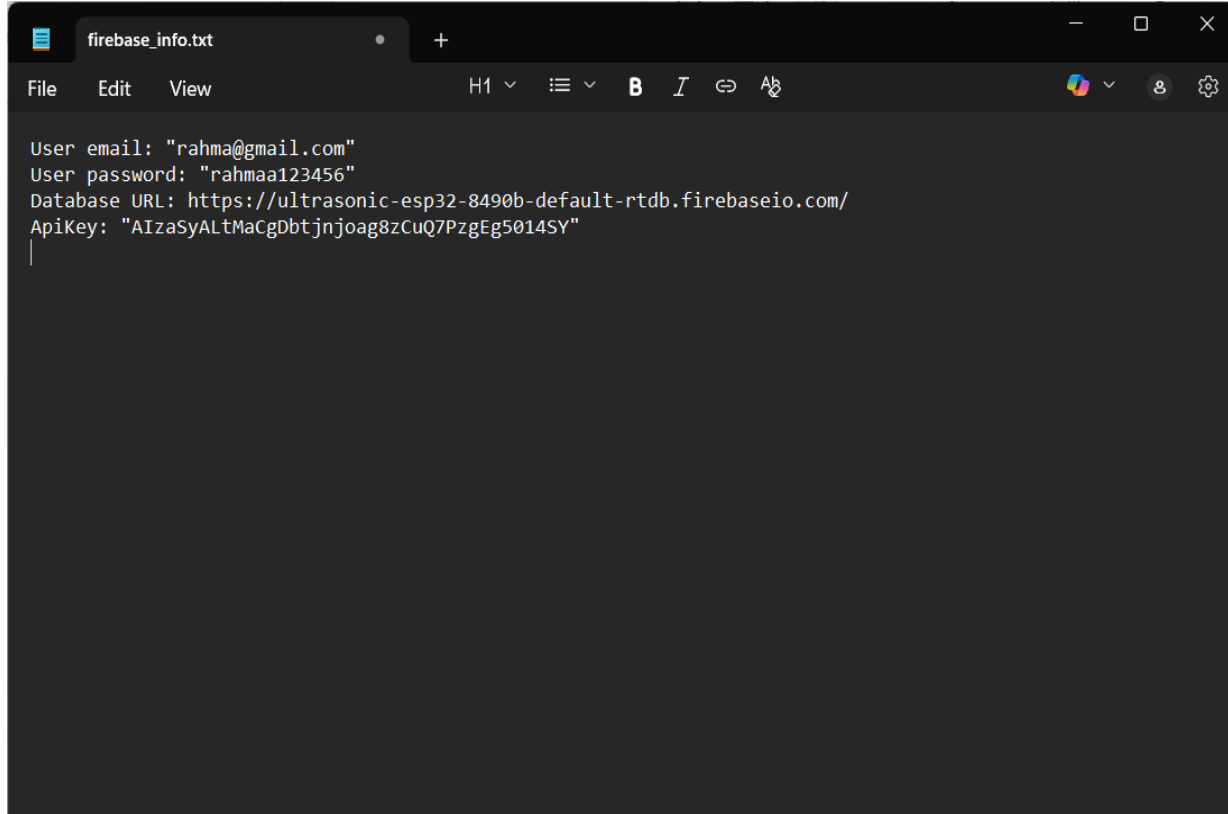
// Your web app's Firebase configuration
// For Firebase JS SDK v7.20.0 and later, measurementId is optional
const firebaseConfig = {
  apiKey: "AIzaSyAltMaCgDbtjnjoag8zCuQ7PzgEg5014SY",
  authDomain: "ultrasonic-esp32-8490b.firebaseio.com",
  databaseURL: "https://ultrasonic-esp32-8490b.firebaseio.com",
  projectId: "ultrasonic-esp32-8490b",
  storageBucket: "ultrasonic-esp32-8490b.firebaseio.com",
  messagingSenderId: "754507425265",
  appId: "1:754507425265:web:5989e03f0078d974781050",
  measurementId: "G-XZ1CR3PLND"
};
```

Go to docs

Activate Windows
Go to Settings to activate Windows.

Firebase: Creating a Realtime Database

❖ Update your Notepad file with your ApiKey to use them later in the ESP32 code.

A screenshot of a Notepad application window titled 'firebase_info.txt'. The window has a dark theme and a standard menu bar with 'File', 'Edit', and 'View'. The text inside the Notepad is as follows:

```
User email: "rahma@gmail.com"  
User password: "rahmaa123456"  
Database URL: https://ultrasonic-esp32-8490b-default-rtdb.firebaseio.com/  
ApiKey: "AIZAyALTMaCgDbtjnjoag8zCuQ7PzgEg5014SY"
```

The cursor is positioned at the end of the last line. The window's title bar includes standard Windows window controls (minimize, maximize, close) and a tab bar with a single tab labeled 'firebase_info.txt'.

Firebase: Creating a Realtime Database

❖ Scroll down and click **continue to consol**.

Then, initialize Firebase and begin using the SDKs for the products you'd like to use.

```
// Import the functions you need from the SDKs you need
import { initializeApp } from "firebase/app";
import { getAnalytics } from "firebase/analytics";
// TODO: Add SDKs for Firebase products that you want to use
// https://firebase.google.com/docs/web/setup#available-libraries

// Your web app's Firebase configuration
// For Firebase JS SDK v7.20.0 and later, measurementId is optional
const firebaseConfig = {
  apiKey: "AIzaSyALtMaCgDbtjnjoag8zCuQ7PzgEg5014SY",
  authDomain: "ultrasonic-esp32-8490b.firebaseio.com",
  databaseURL: "https://ultrasonic-esp32-8490b-default-rtdb.firebaseio.com",
  projectId: "ultrasonic-esp32-8490b",
  storageBucket: "ultrasonic-esp32-8490b.firebaseio.com",
  messagingSenderId: "754507425265",
  appId: "1:754507425265:web:5989e03f0078d974781050",
  measurementId: "G-XZ1CR3PLND"
};

// Initialize Firebase
const app = initializeApp(firebaseConfig);
const analytics = getAnalytics(app);
```

Note: This option uses the [modular JavaScript SDK](#), which provides reduced SDK size.

Learn more about Firebase for web: [Get Started](#), [Web SDK API Reference](#), [Samples](#)

Continue to console

[Go to docs](#)



Activate Windows
Go to Settings to activate Windows.

Firebase with IoT

- ❖ All Firebase Realtime Database **data is stored as JSON objects**.
- ❖ You can think of the database as a **cloud-hosted JSON tree**.
- ❖ Unlike a SQL database, there are **no tables or records**.
- ❖ When you **add data to the JSON tree**, it becomes a **node in the existing JSON structure** with an **associated key**.

```
{  
  "smartHome": {  
    "kitchen": {  
      "light": "off",  
      "temperature": 20  
    },  
    "bedroom": {  
      "light": "on",  
      "temperature"  
      : 18  
    }  
  }  
}
```

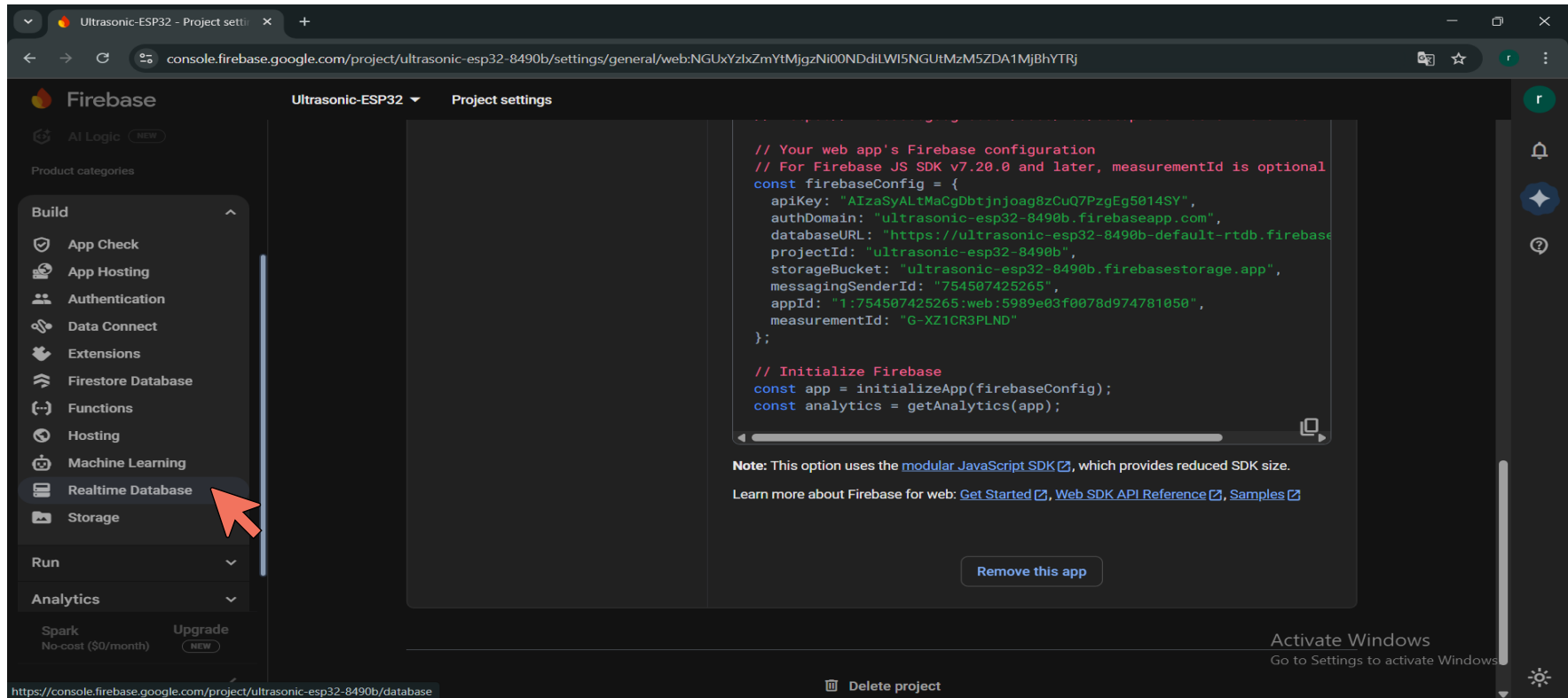


Firebase with IoT

```
{
  "smarthHome": {
    "devices": {
      "light": {
        "status": "on",
        "brightness": 75
      },
      "thermostat": {
        "temperature": 22,
        "mode": "auto"
      },
      "door": {
        "status": "closed"
      }
    },
    "security": {
      "alarm": {
        "status": "armed"
      },
      "camera": {
        "status": "active"
      }
    }
  }
}
```

Firebase: Creating a Realtime Database

❖ Return to your **Realtime Database**.



The screenshot shows the Firebase console interface for a project named "Ultrasonic-ESP32". The left sidebar contains a navigation menu with categories: Build, Run, and Analytics. Under the Build category, "Realtime Database" is highlighted with a red arrow. The main content area displays the "Project settings" for the Realtime Database, including a code editor with the following JavaScript configuration:

```
// Your web app's Firebase configuration
// For Firebase JS SDK v7.20.0 and later, measurementId is optional
const firebaseConfig = {
  apiKey: "AIzaSyALTtMacGDbtjnjoag8zCuQ7PzgEg5014SY",
  authDomain: "ultrasonic-esp32-8490b.firebaseio.com",
  databaseURL: "https://ultrasonic-esp32-8490b-default-rtdb.firebaseio.com",
  projectId: "ultrasonic-esp32-8490b",
  storageBucket: "ultrasonic-esp32-8490b.firebaseio.com",
  messagingSenderId: "754507425265",
  appId: "1:754507425265:web:5989e03f0078d974781050",
  measurementId: "G-XZ1CR3PLND"
};

// Initialize Firebase
const app = initializeApp(firebaseConfig);
const analytics = getAnalytics(app);
```

Below the code editor, a note states: "Note: This option uses the [modular JavaScript SDK](#), which provides reduced SDK size." and a link to "Learn more about Firebase for web: [Get Started](#), [Web SDK API Reference](#), [Samples](#)". A "Remove this app" button is located at the bottom right of the code editor area.

The bottom of the console shows the URL: <https://console.firebase.google.com/project/ultrasonic-esp32-8490b/database> and a "Delete project" button.

Firebase: Creating a Realtime Database

❖ Select **Data** tab then Click on **add Node**.

The screenshot displays the Firebase Realtime Database interface for a project named 'Ultrasonic-ESP32'. The left sidebar contains navigation options: 'Project Overview', 'Project shortcuts', 'Firestore Database', 'Realtime Database' (highlighted in blue), 'What's new', 'AI Logic', 'Product categories', and 'Related development tools'. The main panel shows the 'Realtime Database' section with tabs for 'Data', 'Rules', 'Backups', 'Usage', and 'Extensions'. The 'Data' tab is selected and highlighted with a red box. Below the tabs, a message states: 'Protect your Realtime Database resources from abuse, such as billing fraud or phishing' with a 'Configure App Check' link. The database URL is displayed as 'https://ultrasonic-esp32-8490b-default-rtdb.firebaseio.com'. Below the URL, a text input field contains the same URL, and a red arrow points to a '+' button next to it, indicating the 'add Node' action. The bottom of the screen shows the 'Database location: United States (us-central1)' and an 'Activate Windows' watermark.

Firebase: Creating a Realtime Database

❖ Write your node Key and value choose its datatype and then click Add.

The screenshot displays the Firebase Realtime Database interface for a project named "Ultrasonic-ESP32". The left sidebar shows navigation options: Project Overview, Project shortcuts, Firestore Database, Realtime Database (selected), What's new, AI Logic, and Product categories. The main area shows the Realtime Database URL: `https://ultrasonic-esp32-8490b-default-rtdb.firebaseio.com`. Below the URL, a new node is being added with the key `led` and the value `on`. A dropdown menu is open, showing the data type selection options: Auto, Boolean, Number, Object, and String. A red arrow points to the "String" option, which is highlighted. The interface also includes a "Need help with Realtime Database? Ask Gemini" button, a "Protect your Realtime Database resources from abuse" warning, and a "Database location: United States (us-central1)" indicator at the bottom.

Firebase: Creating a Realtime Database

❖ I added a new node named “distance” and set its value to 10 to represent the ultrasonic sensor reading.

The screenshot displays the Firebase Realtime Database interface for a project named 'Ultrasonic-ESP32'. The left sidebar shows navigation options: Project Overview, Project shortcuts, Firestore Database, Realtime Database (selected), What's new, AI Logic, and Product categories. The main panel shows the Realtime Database structure with a tree view. A new node 'distance' is being added with a value of 10. A red arrow points to the 'Add' button in the context menu. The context menu also shows options for 'Auto', 'Boolean', 'Number', 'Object', and 'String'. The URL bar shows the database URL: `https://ultrasonic-esp32-8490b-default-rtdb.firebaseio.com`. The bottom of the screen shows the database location: 'United States (us-central1)'.

Ultrasonic-ESP32 ▾

Realtime Database

◆ Need help with Realtime Database? Ask Gemini

Data Rules Backups Usage Extensions

Protect your Realtime Database resources from abuse, such as billing fraud or phishing [Configure App Check](#) ✕

`https://ultrasonic-esp32-8490b-default-rtdb.firebaseio.com`

`https://ultrasonic-esp32-8490b-default-rtdb.firebaseio.com/`

distance 10 123 ✕

Cancel Add * Auto

led: "on" Boolean

123 Number

{ } Object

ABC String

Database location: United States (us-central1)

Activate Windows
Go to Settings to activate Windows.

Firebase: Creating a Realtime Database

❖ Now I have two nodes: “led” to control the LED status and “distance” to store the ultrasonic sensor reading.

The screenshot displays the Firebase Realtime Database interface for a project named "Ultrasonic-ESP32". The left sidebar contains navigation options: Project Overview, Project shortcuts, Firestore Database, Realtime Database (selected), What's new, AI Logic, and Product categories. The main panel shows the Realtime Database overview with tabs for Data, Rules, Backups, Usage, and Extensions. A security warning is visible: "Protect your Realtime Database resources from abuse, such as billing fraud or phishing". Below this, a code editor displays a JSON snapshot of the database state:

```
https://ultrasonic-esp32-8490b-default-rtdb.firebaseio.com/  
{  
  "distance": 10,  
  "led": "on"  
}
```

The JSON data is highlighted with a red box. At the bottom of the console, it indicates the "Database location: United States (us-central1)". An "Activate Windows" watermark is present in the bottom right corner.

- ❖ **Now, let's start connecting our Firebase database with our project.**
- I will start building my project, which consists of an **ultrasonic sensor** and an **LED** connected to the **ESP32**.
- The ultrasonic sensor will measure the distance, and the measured value will be sent to **Firebase**.
- Based on the distance value, the LED will turn **ON if distance < 10 or OFF otherwise** automatically.



Project Component we need !

1 ESP-32 Development Board WiFi Bluetooth.

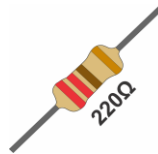
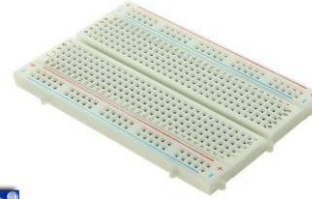
2 breadboard.

3 Ultrasonic Sensor (HC-SR04).

4 Led

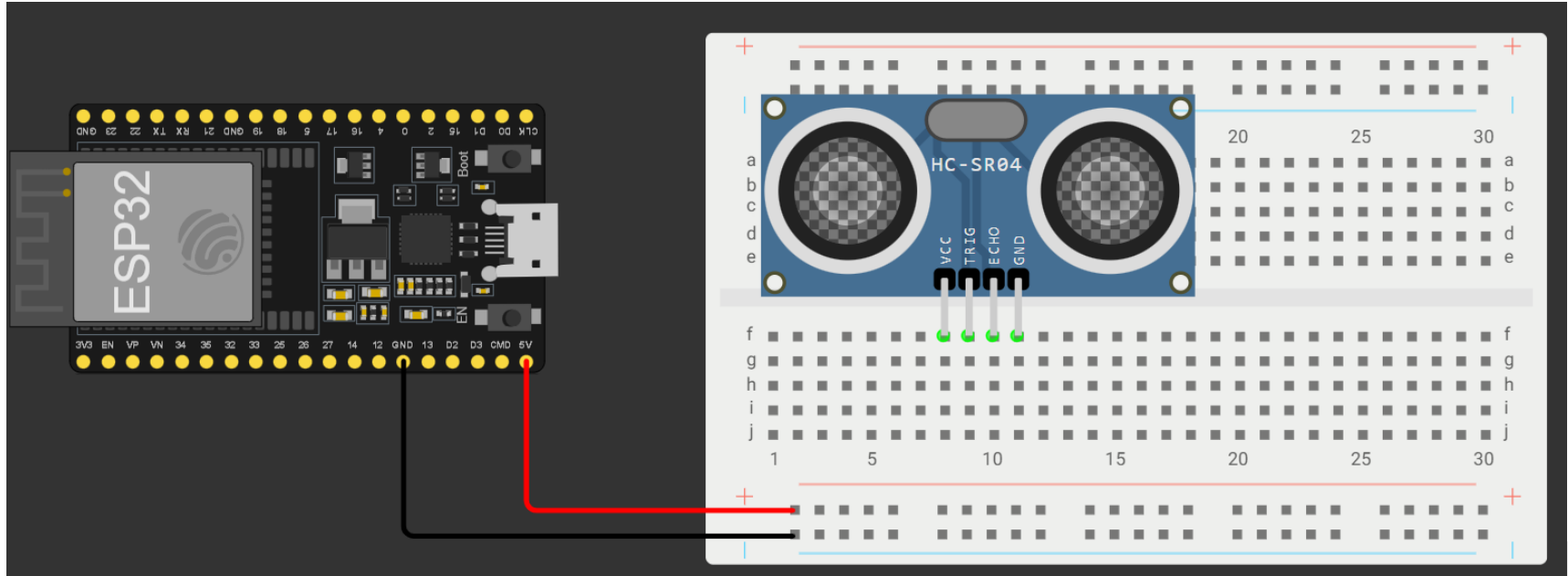
5 Resistor 220ohm

6 Wires



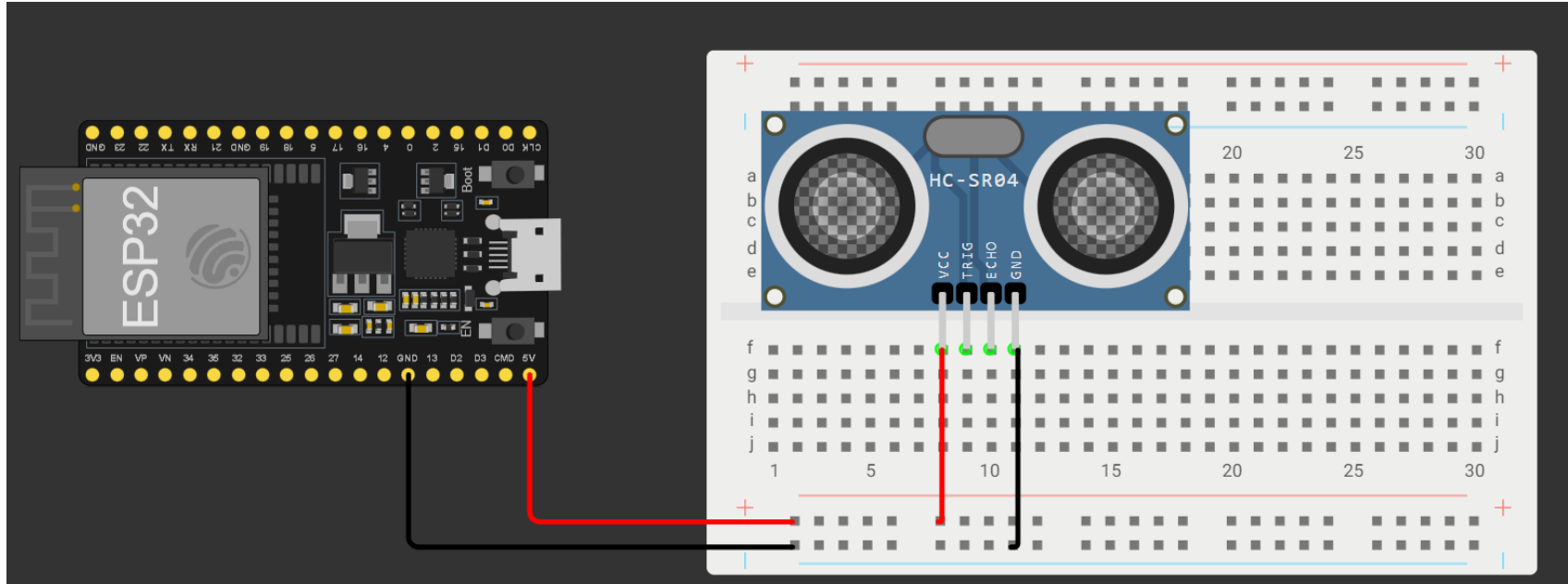
Ultrasonic Sensor with Led

Step1:Connecting the Ultrasonic Sensor to the Breadboard



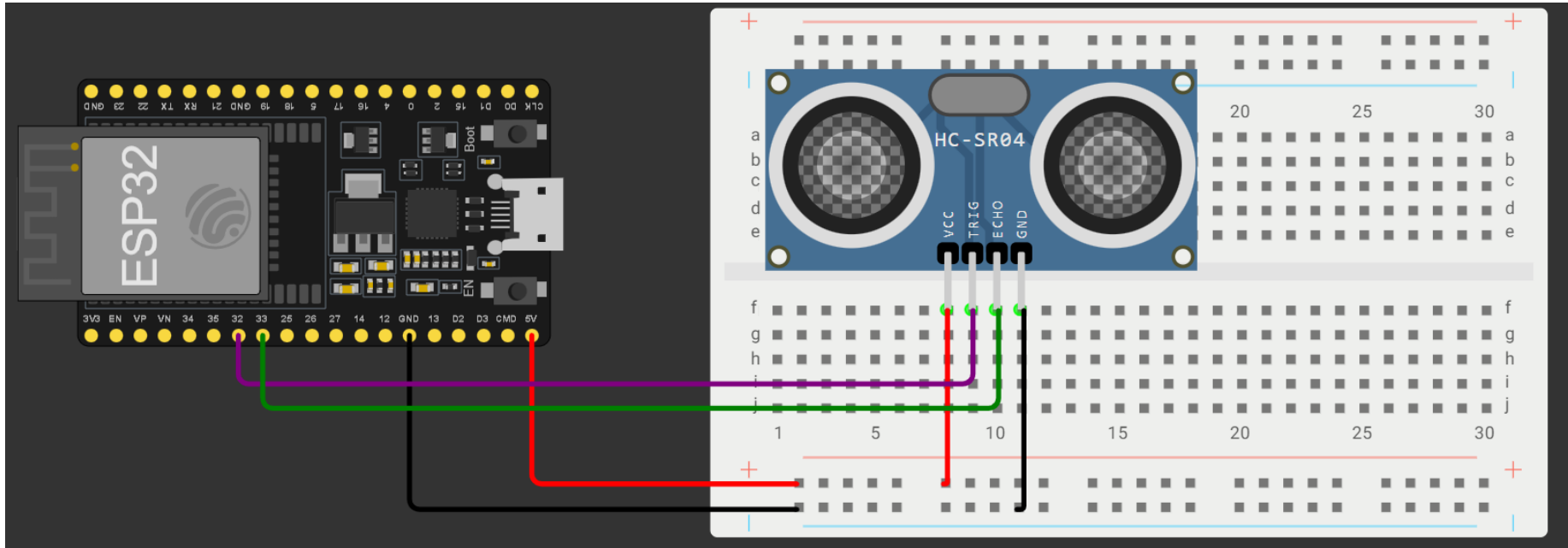
Ultrasonic Sensor with Led

Step 2: Powering the Ultrasonic Sensor



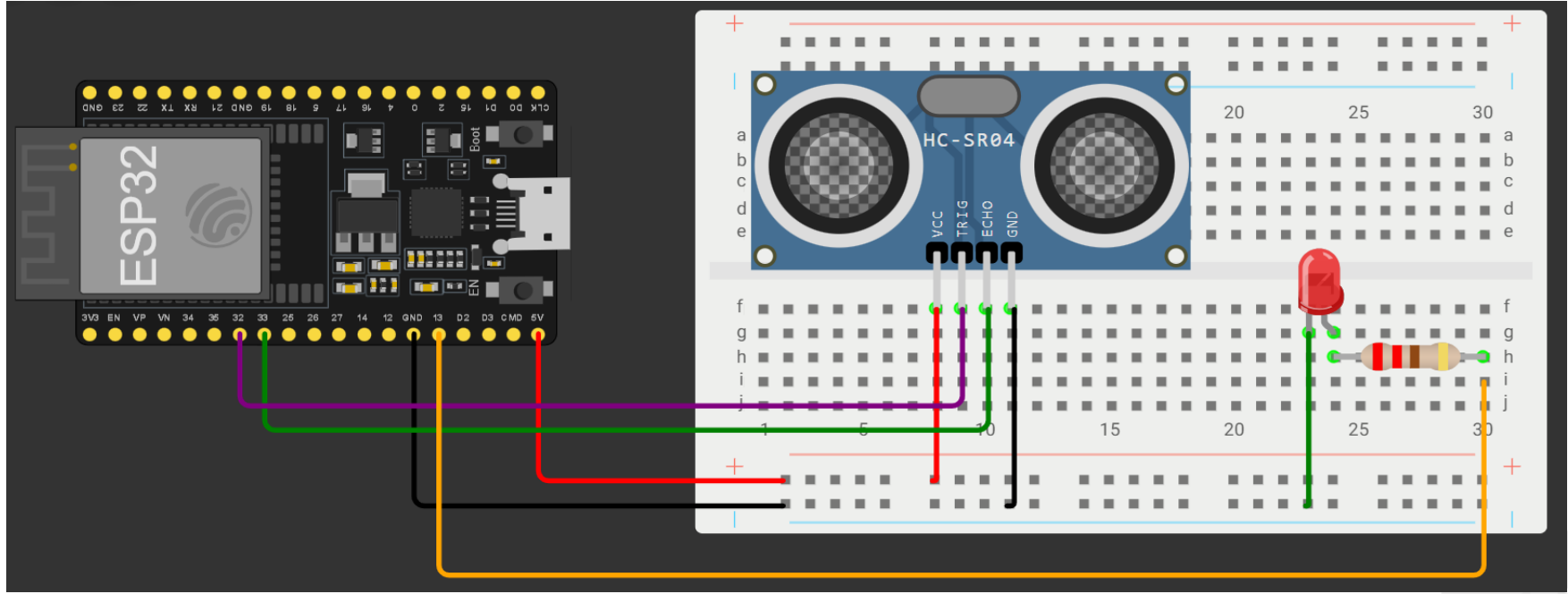
Ultrasonic Sensor with Led

Step 3: Connecting the Control Pins (TRIG and ECHO)



Ultrasonic Sensor with Led

Step 4: Connecting a Led



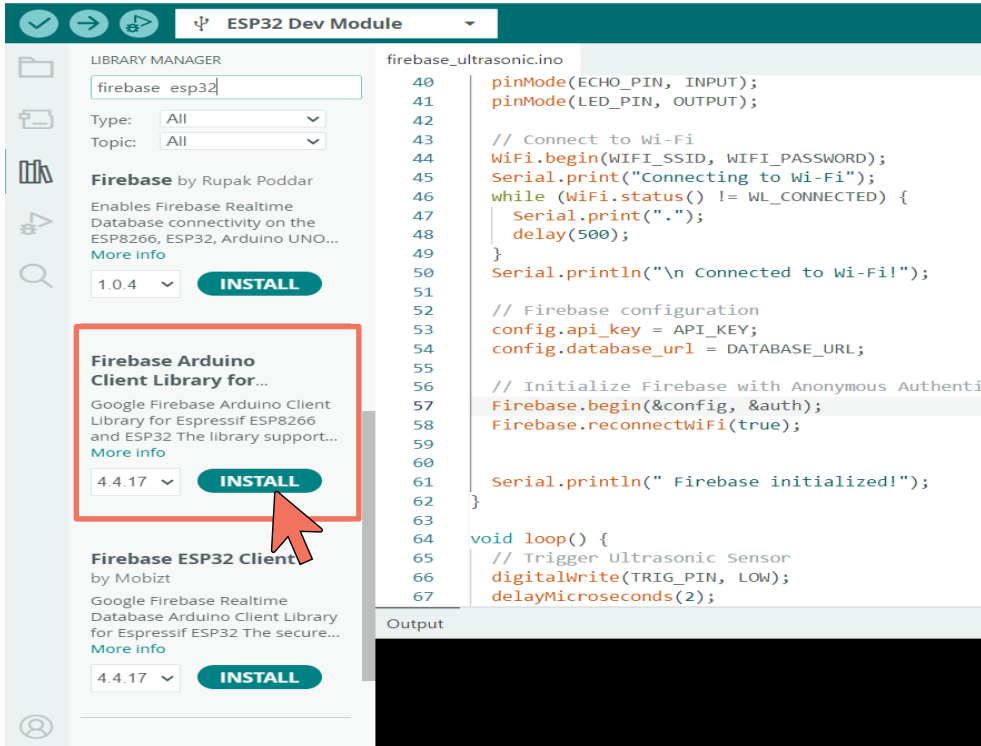
Firebase ESP32 Library

- The **Firebase ESP32** library provides **Firebase Realtime database** and **Firebase Cloud Messaging** functions and **supports ESP32**.



Firebase ESP32 Library

First, we need to install Firebase Library.



The screenshot shows the Arduino IDE interface. At the top, the 'ESP32 Dev Module' is selected. The 'LIBRARY MANAGER' is open, and 'firebase esp32' is entered in the search bar. The search results show two libraries:

- Firebase** by Rupak Poddar: Version 1.0.4, with an 'INSTALL' button.
- Firebase Arduino Client Library for...** by Google: Version 4.4.17, with an 'INSTALL' button. This library is highlighted with a red box, and a red arrow points to its 'INSTALL' button.
- Firebase ESP32 Client** by Mobitz: Version 4.4.17, with an 'INSTALL' button.

The code editor on the right shows the 'firebase_ultrasonic.ino' file with the following code:

```
40 pinMode(ECHO_PIN, INPUT);
41 pinMode(LED_PIN, OUTPUT);
42
43 // Connect to Wi-Fi
44 WiFi.begin(WIFI_SSID, WIFI_PASSWORD);
45 Serial.print("Connecting to Wi-Fi");
46 while (WiFi.status() != WL_CONNECTED) {
47   Serial.print(".");
48   delay(500);
49 }
50 Serial.println("\n Connected to Wi-Fi!");
51
52 // Firebase configuration
53 config.api_key = API_KEY;
54 config.database_url = DATABASE_URL;
55
56 // Initialize Firebase with Anonymous Authent
57 Firebase.begin(&config, &auth);
58 Firebase.reconnectWiFi(true);
59
60
61 Serial.println(" Firebase initialized!");
62
63
64 void loop() {
65   // Trigger Ultrasonic Sensor
66   digitalWrite(TRIG_PIN, LOW);
67   delayMicroseconds(2);
```

The 'Output' window at the bottom is empty.

1. go to library manager from **tools** → **manage libraires** and search for (firebase esp32)
2. Make sure to install the library shown in this picture.





ESP32 Dev Module

firebase_ultrasonic.ino

```
1  #include <WiFi.h>           // Library to connect ESP32 to Wi-Fi
2  #include <Firebase_ESP_Client.h> // Library for Firebase functions
3  #include "addons/TokenHelper.h" // Required helper for Firebase authentication
4  #include "addons/RTDBHelper.h" // Required helper for Firebase Realtime Database
5
6  // Wi-Fi credentials
7  #define WIFI_SSID "rahmabadar"
8  #define WIFI_PASSWORD "rahma@011"
9
10 // Firebase credentials
11 #define API_KEY "AIzaSyALTMaCgDbtjnjoag8zCuQ7PzgEg5014SY"
12 #define DATABASE_URL "https://ultrasonic-esp32-8490b-default-rtdb.firebaseio.com/"
13 #define USER_EMAIL "rahma@gmail.com"
14 #define USER_PASSWORD "rahmaa123456"
15
16 // Pin Definitions
17 #define TRIG_PIN 32 // Trigger pin of ultrasonic sensor
18 #define ECHO_PIN 33 // Echo pin of ultrasonic sensor
19 #define LED_PIN 13 // Built-in LED pin
20
21 // Firebase Objects
22 FirebaseData fbdo; // Object to handle Firebase data
23 FirebaseAuth auth; // Object for Firebase authentication
24 FirebaseConfig config; // Object for Firebase configuration
25
26 // Variables for distance measurement
27 long duration;
28 float distance;
29
```



firebase_ultrasonic.ino

```
30 void setup() {
31   Serial.begin(115200); // Start serial communication for debugging
32
33   // Initialize pins
34   pinMode(TRIG_PIN, OUTPUT);
35   pinMode(ECHO_PIN, INPUT);
36   pinMode(LED_PIN, OUTPUT);
37
38   // Connect to Wi-Fi
39   WiFi.begin(WIFI_SSID, WIFI_PASSWORD);
40   Serial.print("Connecting to Wi-Fi");
41   while (WiFi.status() != WL_CONNECTED) {
42     Serial.print(".");
43     delay(500);
44   }
45   Serial.println("\nConnected to Wi-Fi!");
46
47   // Configure Firebase
48   config.api_key = API_KEY;
49   config.database_url = DATABASE_URL;
50
51   auth.user.email = USER_EMAIL;
52   auth.user.password = USER_PASSWORD;
53
54   // Initialize Firebase connection
55   Firebase.begin(&config, &auth);
56   Firebase.reconnectWiFi(true); // Auto-reconnect Wi-Fi if disconnected
57
58   Serial.println("Firebase initialized!");
59 }
60
```



firebase_ultrasonic.ino

```
61 void loop() {
62   // Measure distance using ultrasonic sensor
63   digitalWrite(TRIG_PIN, LOW);
64   delayMicroseconds(2);
65   digitalWrite(TRIG_PIN, HIGH);
66   delayMicroseconds(10);
67   digitalWrite(TRIG_PIN, LOW);
68
69   duration = pulseIn(ECHO_PIN, HIGH); // Measure the time for echo to return
70   distance = duration * 0.034 / 2; // Convert time to distance in cm
71
72   Serial.print("Distance: ");
73   Serial.println(distance);
74
75   // Send distance to Firebase
76   if (Firebase.RTDB.setFloat(&fbdo, "/distance", distance)) {
77     Serial.println("Distance updated in Firebase");
78   } else {
79     Serial.print("Failed to send distance: ");
80     Serial.println(fbdo.errorReason());
81   }
82   // Control LED based on distance
83   if (distance < 10) {
84     digitalWrite(LED_PIN, HIGH); // Turn LED ON if object is closer than 10 cm
85     Firebase.RTDB.setString(&fbdo, "/led", "ON");
86   } else {
87     digitalWrite(LED_PIN, LOW); // Turn LED OFF otherwise
88     Firebase.RTDB.setString(&fbdo, "/led", "OFF");
89   }
90   delay(2000); // Wait 2 seconds before next measurement
91 }
```

Ultrasonic Sensor with Led Data stored in firebase

firebase_ultrasonic.ino

```
1 #include <WiFi.h>           // Library to connect ESP32 to Wi-Fi
2 #include <Firebase_ESP_Client.h> // Library for Firebase functions
3 #include "addons/TokenHelper.h" // Required helper for Firebase authentication
4 #include "addons/RTDBHelper.h" // Required helper for Firebase Realtime Database
5
6 // Wi-Fi credentials
7 #define WIFI_SSID "rahmabadar"
8 #define WIFI_PASSWORD "rahma@011"
```

Explanation:

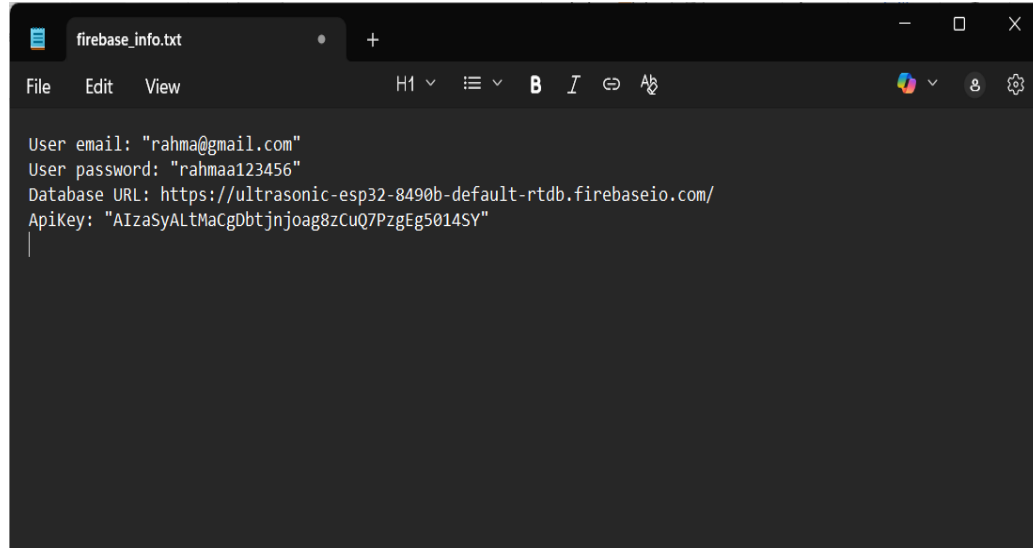
- Here we enter the **Wi-Fi network name (SSID)**
- and **password** so that the **ESP32** can connect to the internet.

This connection allows the board to send and receive data from **Firebase**



Ultrasonic Sensor with Led Data stored in firebase

```
10 // Firebase credentials
11 #define API_KEY "AIzaSyAltMacGDbtjnjoag8zCuQ7PzgEg5014SY"
12 #define DATABASE_URL "https://ultrasonic-esp32-8490b-default-rtdb.firebaseio.com/"
13 #define USER_EMAIL "rahma@gmail.com"
14 #define USER_PASSWORD "rahmaa123456"
```



A screenshot of a Notepad application window. The title bar shows 'firebase_info.txt'. The menu bar includes 'File', 'Edit', and 'View'. The toolbar shows various editing icons. The text area contains the same code as the previous block, with line numbers 10 through 14. The text is as follows:

```
User email: "rahma@gmail.com"
User password: "rahmaa123456"
Database URL: https://ultrasonic-esp32-8490b-default-rtdb.firebaseio.com/
ApiKey: "AIzaSyAltMacGDbtjnjoag8zCuQ7PzgEg5014SY"
```

Explanation:

- Here is where we put the information we saved earlier in **Notepad** while setting up the project.

These include:

- **API_KEY** → the unique key for our Firebase project.
- **DATABASE_URL** → the address of our Realtime Database.
- **USER_EMAIL & USER_PASSWORD** → the login details we created for Firebase.



firebase_ultrasonic | Arduino IDE 2.3.6

File Edit Sketch Tools Help

ESP32 Dev Module

```
firebase_ultrasonic.ino
1 #include <WiFi.h> // Library to use WiFi
2 #include <Firebase_ESP_Client.h> // Library for Firebase Realtime Database

Output Serial Monitor X

Message (Enter to send message to 'ESP32 Dev Module' on 'COM4')

..
Connected to Wi-Fi!
Firebase initialized!
Distance: 804.70
Distance updated in Firebase
Distance: 5.58
Distance updated in Firebase
Distance: 5.83
Distance updated in Firebase
Distance: 5.58
Distance updated in Firebase
Distance: 5.01
Distance updated in Firebase
Distance: 6.85
Distance updated in Firebase
Distance: 5.01
Distance updated in Firebase
```

Ultrasonic-ESP32 - Realtime Database

console.firebase.google.com/project/ultrasonic-esp32-8490b/database/ultrasonic-esp32-8490b-default-rtdb/data

Realtime Database

Need help with Realtime Database? Ask Gemini

Data Rules Backups Usage Extensions

Protect your Realtime Database resources from abuse, such as billing fraud or phishing [Configure App Check](#)

<https://ultrasonic-esp32-8490b-default-rtdb.firebaseio.com/>

```
{
  "distance": 5.015,
  "led": "ON"
}
```

Database location: United States (us-central1)

Activate Windows
Go to Settings to activate Windows.

```

ESP32 Dev Module
firebase_ultrasonic.ino
1 #include <WiFi.h> // Library to
2 #include <Firebase_ESP_Client.h> // Library for
Output Serial Monitor X
Message (Enter to send message to 'ESP32 Dev Module' on 'COM4')
Distance updated in Firebase
Distance: 19.74
Distance updated in Firebase
Distance: 19.47
Distance updated in Firebase
Distance: 24.04
Distance updated in Firebase
Distance: 21.96
Distance updated in Firebase
Distance: 804.73
Distance updated in Firebase
Distance: 21.03
Distance updated in Firebase
Distance: 21.03
Distance updated in Firebase
Distance: 804.59
Distance updated in Firebase
Distance: 804.66
Distance updated in Firebase
Distance: 804.59
Distance updated in Firebase
Distance: 21.49
Distance updated in Firebase
Distance: 804.56
Distance updated in Firebase
Distance: 19.47
Distance updated in Firebase
Distance: 19.47
Distance updated in Firebase
Distance: 21.03
Distance updated in Firebase

```

Firebase

Project shortcuts

- Firestore Database
- Realtime Database**
- Authentication

What's new

- AI Logic NEW

Product categories

- Build
- Run
- Analytics
- AI

Related development tools

- Firebase Studio

Spark No-cost (\$0/month) Upgrade NEW

Realtime Database

[Need help with Realtime Database? Ask Gemini](#)

[Data](#) [Rules](#) [Backups](#) [Usage](#) [Extensions](#)

[Protect your Realtime Database resources from abuse, such as billing fraud or phishing](#) [Configure App Check](#)

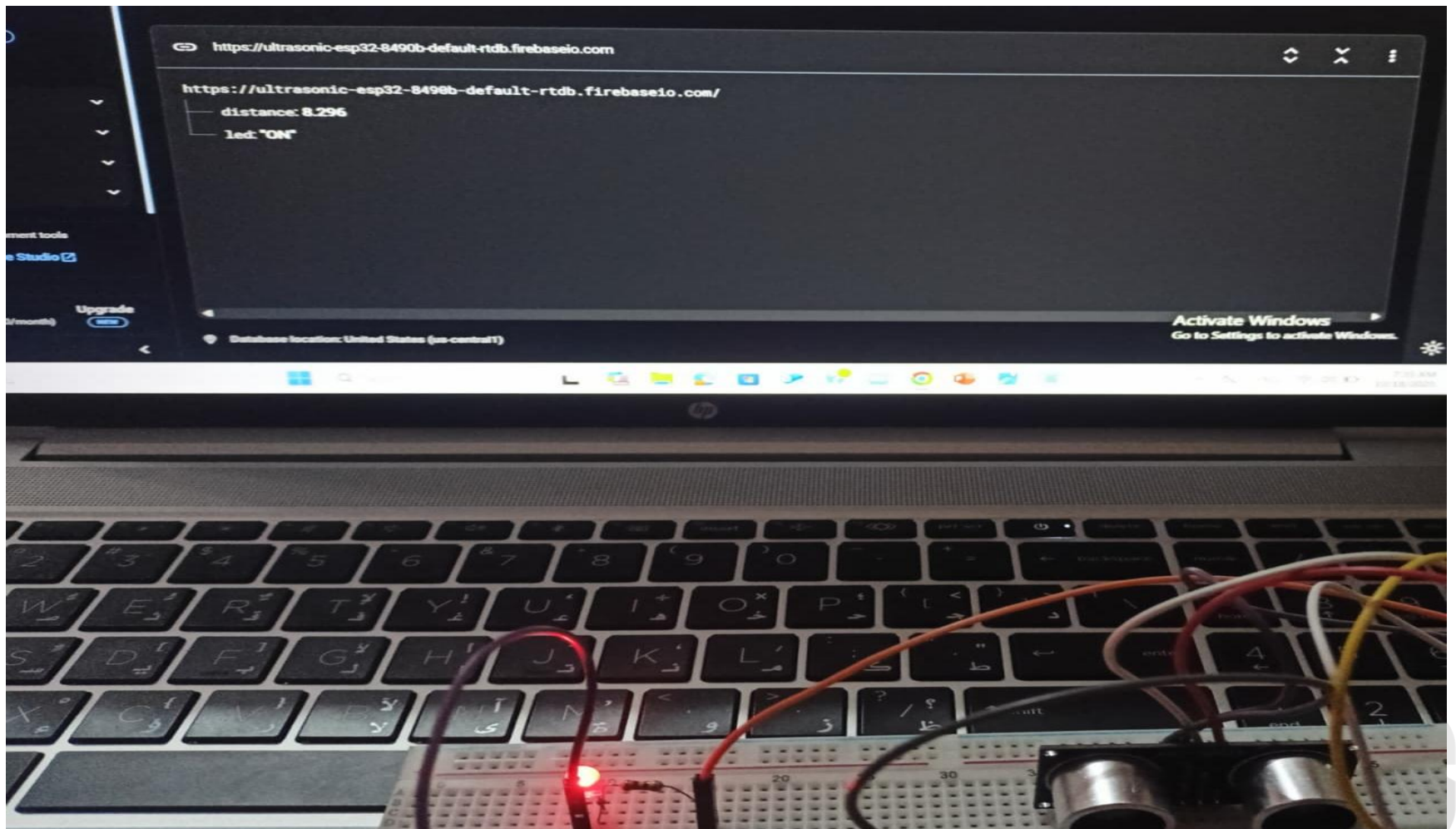
<https://ultrasonic-esp32-8490b-default-rtdb.firebaseio.com>

```

https://ultrasonic-esp32-8490b-default-rtdb.firebaseio.com/
{
  distance: 21.029
  led: "OFF"
}

```

Database location: United States (us-central1)



Thank you

